

Quantum

Group Lead • Ridgeway Quantum Laboratory

15 DAYS • 61 STOPS • GRADE — • QUANTUM

THE OPENING CARD

Ridgeway runs a quantum processor: twelve qubits, held a hundredth of a degree above absolute zero by a refrigerator the size of a wardrobe. Everyone calls it the fridge. You lead the group. A laboratory in Delft has published a result your funder has already asked about. Your own best number was measured on one machine, by one group, in one week. The review board sits in a fortnight. This morning the fridge came down to forty-two thousandths of a degree instead of eleven. It sounds like a small miss. It is the difference between a quantum processor and a warm chip. Priya Raghavan needs to know which you have, and she needs it before the board does.

Cold enough to have a ground state

BEFORE THE DAY — GREET

The spine, and everybody working along it

The laboratory is one corridor from the warm end to the cold, and the people along it barely overlap: fabrication in the morning, control at all hours, verification whenever the fridge is cold. Raghavan, the group lead, wants you known to as many of them as you can in one pass, because a chip is handed between four of these people before anybody measures anything.

PLAN CARD 114W

Monday, and the machine is warm. Ijeoma Okafor brought the fridge down over the weekend and it has settled at 42 millikelvin instead of the 11 it usually holds, which sounds like a small difference and is not. Kofi Mensah cannot get a clean measurement out of any qubit on the chip, and Priya Raghavan has a review in ten days. Okafor thinks the heat is coming down a cable somebody added in the last cooldown. Today you work out what temperature actually does to a qubit, put a number on how much of the chip is sitting in the wrong state before anybody touches it, and find where the heat is getting in.

BRIEFING 15W

The fridge is at 42 millikelvin instead of 11, and nothing on the chip behaves.

WHAT YOU DECIDE 15W

Work out why the chip has to be colder than anything else in the building.

1.1 Which stage the leak is on

CRYOGENICS & VACUUM · A ROOM · PROBE

SCENE 61W

A dilution refrigerator cools in stages. Each one is colder than the stage above it: 50 kelvin, 4 kelvin, the still, the heat exchanger, the cold plate, and the mixing chamber at the bottom. Heat travels down that chain and stops where it is anchored.

Ridgeway's fridge has stalled warm. Okafor has last cooldown's numbers and a thermometer on every stage.

GUIDE 75W

Nothing has been measured for you. Press Read on any stage and it reports what it holds now, what it held last cooldown, and how hard its cooling is working. Take at least three, and remember that a stage matching last time tells you as much as one that does not. Then click the stage where the heat is arriving and commit. You are graded on the stage, not on how many readings it took.

ASK 28W

The bottom of the fridge is warm and the top may not be. Take the readings you think you need — at which stage is the heat arriving?

BEHIND THE BUTTON 196W

How the stages work. Each stage is anchored to the one above by a deliberate thermal link, and cools whatever is bolted to it. A stage removes a fixed amount of heat per second, so extra heat arriving does not simply pass through: the stage warms and its cooling works harder, and everything below it is left with less cooling to spare. That is why a load shows up as a pattern down the chain rather than at one thermometer.

What the numbers mean. "Holding now" against "last cooldown" is a comparison with a known-good run, which is worth more here than any absolute temperature — a stage sitting at its usual value under its usual load has been ruled out, and one sitting high has not. The cooling effort is the third column: a stage can hold its temperature and still be telling you something, if it is working harder to do it.

Why the top of the chain matters. A vacuum leak, a failing pulse tube, or a room-temperature fault all deliver heat to the warmest stages first. If those are unchanged, a whole family of explanations is gone before you look further down.

TAKES AS READ

a refrigerator removes heat in stages, each colder than the last · a conductive path carries heat to wherever it is anchored, and past wherever it is not · thermal energy against level spacing — why a qubit needs millikelvin — taken as read

VERDICT 158W

Two stages held exactly what they held last time, which is what rules almost everything out: a vacuum problem, a failing pulse tube, a room-temperature fault would all warm the top of the chain first, and the top of the chain has not moved. From the still downwards every stage is high and working harder, and heat only

behaves that way when it is being delivered from above. So the deepest stage is the worst affected and is not where the problem is — the load enters at the first stage that departs, and the stages below it are inheriting it. That is a conductive path anchored at 50 K and 4 K and not below, which is exactly what a line added since the last cooldown and heat-sunk at only the easy stages does. Okafor's cable is now worth looking at, not because cables are always guilty but because this pattern is what an unanchored one predicts.

TAKEAWAY 17W

A refrigerator's stage-by-stage temperature pattern can localise a heat load instead of merely announcing that one exists.

1.2 The energy of one quantum against the energy of the room

CRYOGENICS & VACUUM · A PERSON · BALLPARK

SCENE 38W

The chip's qubits answer at about 5 gigahertz. Okafor circles the Delft date on the whiteboard and writes two energy scales beside it: one quantum at that frequency, and the thermal energy at 42 millikelvin. Then she waits.

GUIDE 75W

Five numbers, and three of them are energies while two are the raw frequency and temperature those energies were built from. Two of the energies are thermal, taken at different temperatures. The exponent is a ratio of energies, so both terms have to be in joules and both have to describe today's fridge. Using the colder thermal energy answers a question about a machine that is working. That is not the one on the bench.

ASK 13W

Using the 2 energy scales, estimate the equilibrium excited-state fraction at 42 mK.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a system in contact with something warm can absorb energy from it

VERDICT 91W

"A qubit sitting in a warm environment does not stay in its ground state: it is being offered energy all the time, and it takes some. The fraction that ends up excited follows the ratio of the thermal energy to the energy of one quantum, exponentially, which is why a factor of four in temperature is not a factor of four in behaviour. At 11 millikelvin almost nothing is excited. At 42 the population is large enough that a measurement of \"the ground state\" is partly a measurement of the fridge."

TAKEAWAY 17W

Thermal energy well below one quantum is what keeps a qubit in its ground state at rest.

1.3 What is worth doing at the wrong temperature

CONTROL & READOUT · A ROOM · TRIAGE

SCENE 37W

Fixing the heat leak means a warm-up and another cooldown: four days. With Delft's slot approaching, Mensah does not want four empty days and asks what work remains interpretable while the device is off its normal temperature.

GUIDE 63W

All four options are things the group could genuinely do in four days. They differ in whether the number that comes out still means what it usually means. Sort them by what the answer depends on: the chip, the wiring, or both. A measurement taken off-nominal is not useless. A warm-state diagnostic filed as a baseline is worse than no measurement at all.

ASK 19W

What is the best use of the off-nominal period if Ridgeway needs trustworthy baseline numbers before the Delft run?

BEHIND THE BUTTON 194W

Why first is a different question from most important. When several calls compete for the same hour, what a choice is worth is not its own importance but the difference between doing it now and doing it later. A serious problem that will be no worse in an hour costs nothing to defer. A smaller one that closes a door costs everything behind that door.

What to look for in the options. Two things separate them: which is still changing, and which is a precondition for the others. A situation that is deteriorating has a cost per hour attached to it, and a step that unblocks the rest multiplies the value of the hours after it. Everything else is a preference about where to start.

Why only one answer is marked. In the situation these options describe all of them eventually happen; what is being tested is the head of the queue, because that is where the reasoning is visible. The verdict names what each of the others was waiting on, which is worth reading even when the choice was right — the ordering behind the first place is the rest of the answer.

TAKES AS READ

some measurements depend on temperature and some do not · energy quantisation, and why a level spacing sets everything else — taken as read

VERDICT 93W

The machine is not useless at 42 millikelvin, but the question attached to each measurement changes. Thermal population can be measured precisely, and temperature-dependent T1, readout and resonator occupancy can diagnose what the heat is doing. Those are useful *warm-state diagnostics*, not the nominal figures the review needs. The best use of the downtime is therefore work whose interpretation survives the off-nominal temperature — wiring continuity, attenuation, gain and timing — plus explicitly labelled temperature diagnostics. Review-quality coherence,

readout and benchmarking should be repeated after the fridge returns to its normal operating point.

TAKEAWAY 18W

An off-nominal machine can still produce useful diagnostics, but those data must not be mistaken for nominal performance.

END OF THE DAY 17W

A qubit is only a two-state system while the room around it has no energy to spare.

Finding a qubit that will not answer

BEFORE THE DAY — TRIAL

Walk the spine, warm end to cold

The corridor is a temperature gradient and the order of the rooms along it is the order the work happens in. Walk it once now. Everything in this building costs time in proportion to how far along the spine it is, and a plan that ignores that spends the day going back and forth past the same doors.

PLAN CARD 105W

Friday, the fridge is at 11 millikelvin and 11 of the 12 qubits are where they were last month. The twelfth is not answering anywhere near its design frequency, and Kofi Mensah has spent the morning sweeping. Sara Lindqvist thinks the junction on that one came out of a run she has been suspicious of since the spring. Priya Raghavan needs 12 working qubits for the circuit she has promised the review. Today you find the transition by sweeping for it, turn the drive into a pulse that flips the qubit reliably, and decide whether this chip is the one to spend the fortnight on.

BRIEFING 17W

The chip is cold again and one qubit is not where the design says it should be.

WHAT YOU DECIDE 13W

Locate a transition and turn a drive into a pulse that flips it.

2.1 Where the qubit actually is

CONTROL & READOUT · A ROOM · SWEEP

SCENE 57W

A qubit absorbs a drive only near its own transition frequency. So you find that frequency by sweeping the drive and watching how much of the population ends up excited. The design put this one at 5.02 gigahertz. The readout there has been flat all morning, so the device that came out of fabrication is somewhere else.

GUIDE 69W

Drag the drive frequency. The readout is the fraction of shots that finish excited, so a frequency this qubit absorbs shows up as a rise, and only the places you stop at get plotted. Sweep the whole band before deciding: more than one feature can appear, and they are not the same size. What counts as the transition is on the Background button. Then mark the frequency and commit.

ASK 15W

Nothing responds at the design frequency. Which frequency does this qubit absorb the drive at?

BEHIND THE BUTTON 170W

Why the design number is only a guess. The transition frequency is set by a junction barrier about a nanometre thick, and no fabrication process controls a nanometre exactly. A device landing a few hundred megahertz from its design value is ordinary, which is why every qubit is found by spectroscopy before it is used.

What a resonant drive does. On its own transition, a drive moves nearly the whole population into the excited state, because every repetition is driving the same transition in step. Off resonance the population barely moves. So the size of a feature is evidence about what it is, not merely about how strong the drive was.

What else can absorb. A smaller feature elsewhere in the band is usually real and usually not the qubit: a two-level defect in the oxide, a stray mode of the package, or a neighbouring qubit. Those are worth knowing about — one of them turns up again later this fortnight as a place this qubit must not be parked.

TAKES AS READ

a system absorbs energy strongly only near its own transition frequency

VERDICT 108W

A qubit absorbs a drive only close to its own transition, so the frequency where the population goes up is the transition. It is nowhere near 5.02 GHz because that number was a prediction: the frequency depends on a junction barrier about a nanometre thick, and no process controls a nanometre exactly. Two responses appear, and size is what separates them — a transition driven on resonance takes almost the whole population, while the small feature takes a sixth of it and is something else on the chip. Which of the two is the qubit can be settled by driving harder and watching only one of them saturate.

TAKEAWAY 16W

A designed transition frequency is a target; spectroscopy tells you where the fabricated device actually responds.

2.2 How long a flip takes

CONTROL & READOUT · A PERSON · BALLPARK

SCENE 23W

With the drive at 4.61 GHz, the qubit's population oscillates as the pulse gets longer. Mensah measures a full cycle every 84 nanoseconds.

GUIDE 77W

Five numbers and only two of them describe the oscillation. The rest are the carrier frequency, the temperature and the size of the chip. All are true of the device; none is about how long a rotation takes. Ask of each whether it describes the population moving or the machine it moves in. Confusing the drive frequency with the rate the population turns at is the standard error here, and the two differ by orders of magnitude.

ASK 10W

Estimate the length of a pulse that flips this qubit.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a driven two-level system cycles between its states while the drive is on ·
superposition as an amplitude, not a mixture — taken as read · resonance, detuning,
and finding a qubit frequency — taken as read

VERDICT 72W

"Driving on resonance does not simply leave the qubit excited. Under a coherent drive the population oscillates between the two basis states at the Rabi frequency. A π pulse corresponds to half of 1 population cycle, so the 84 ns period gives a first calibration near 42 ns. That number is only a starting point: changing drive amplitude changes the Rabi frequency, and strong drives can introduce leakage or other off-resonant effects."

TAKEAWAY 17W

A resonant drive produces an oscillation; the pulse area, not simply the carrier frequency, determines the rotation.

2.3 1 bad qubit out of 12

FABRICATION & MATERIALS · A ROOM · SCIENCETANK

SCENE 40W

The twelfth qubit is found, but the weak feature beside it survives every repeat. Swapping chips would consume four of the days before Delft's slot. Lindqvist arrives with the fabrication coupon: its junction resistances spread twice as wide as usual.

GUIDE 88W

The chip has twelve qubits and eleven sit within 40 MHz of design. A warm-up and cooldown is four days out of the ten before the review. The spare comes from the same suspicious run and its coupon has not been measured. Junction resistance on the coupon spreads twice as wide as usual. The defect near 4.55 GHz has not been characterised on any chip in this laboratory. A register reported as eleven qubits is a different claim from one reported as twelve with one of them worse.

ASK 6W

Decide whether to keep this chip

BEHIND THE BUTTON 193W

Why the whole spread is graded. Funding is not a vote for one idea. A portfolio says what you think is likely, what is worth hedging against, and what is not worth pursuing at all — and the last two are where most of the information is. Backing the right proposal while quietly funding a bad one is a worse answer than it looks. That is why the small numbers count as much as the big one.

What the three numbers are for. Thirty-five is what makes a lead a lead: below it you have hedged rather than chosen. Fifteen is the most that can sit on unsupported work before it stops being a rounding error. Past that it is a second opinion nobody argued for. Twenty is the floor under a line of work you have already called strong, because funding it too thin to finish spends the money and buys nothing.

Why there is a floor on the total. Points held back are not caution; they are a decision not to decide, taken with somebody else's money and somebody else's deadline. The floor is what forces the panel to say something.

TAKES AS READ

a fabrication run produces chips with a spread of properties · energy quantisation, and why a level spacing sets everything else — taken as read

VERDICT 103W

The useful question is not whether this chip is perfect; it is whether changing chips is likely to improve the run Ridgeway is about to send to Delft. The present chip has 11 qubits close to target and a twelfth that can be driven and characterised. A swap costs four days, and the spare comes from the same suspicious fabrication run, so replacement does not remove the underlying uncertainty. The strongest plan is to keep this chip, map the nearby resonance, operate the qubit away from it if possible, and give Lindqvist the data needed to decide whether the feature tracks the process.

TAKEAWAY 19W

A device decision should compare the cost of its known imperfection with the cost and uncertainty of replacing it.

2.4 Ten millikelvin, while the wiring warms

CRYOGENICS & VACUUM · A ROOM · HOLD

SCENE 38W

The chip has to sit at ten millikelvin while a measurement runs. A new coaxial line is carrying heat down from room temperature, the still is working, and the flow control is the only thing that answers it.

GUIDE 49W

Hold the mixing chamber temperature inside the band on the thermometry. The band narrows through the run, because the same excess temperature costs more the longer the qubit has to stay coherent. The flow control is your control, and every heat load keeps arriving until the circulation answers it.

ASK 11W

Hold the mixing chamber at ten millikelvin while the measurement runs.

BEHIND THE BUTTON 129W

Why ten millikelvin and not one kelvin. A qubit's two levels are a few gigahertz apart, and thermal energy at one kelvin is comparable to that spacing — so the qubit sits in a mixture rather than in its ground state, and every experiment starts from noise.

Why a heat load is a rate. A coaxial line conducting from room temperature does not raise the chamber by a millikelvin and stop; it carries heat continuously, so the temperature climbs until the circulation carries the same power away.

Why the margin tightens. The longer a measurement runs, the more of the qubit's coherence has already been spent, so a warm spell late in a run costs a larger share of what was left than the same excursion at the start.

TAKES AS READ

a system in thermal contact settles toward the temperature around it

VERDICT 161W

A superconducting qubit needs millikelvin because its two levels are only a few gigahertz apart, and thermal energy at a kelvin is comparable to that spacing — the qubit would sit in a thermal mixture rather than in its ground state, and every shot would start from noise. Holding that temperature is a continuous job because every disturbance is a rate. A coaxial line conducts heat down from room temperature for as long as it is installed; a pulse tube cycling changes the load on the still continuously; a helium level falling changes the cooling power available. None of them is a step, so the flow is set to a new rate and left there rather than nudged. The band narrows for the same reason the whole experiment is timed: coherence is being spent as the run goes on, and a warm spell in the last minutes costs a larger share of what remains than the same excursion in the first.

TAKEAWAY 13W

A heat load that keeps arriving is answered by circulation, not by waiting.

END OF THE DAY 24W

A qubit is not at the frequency you designed; it is at the frequency it answers to, and finding that is the first measurement.

Two clocks, and the one that runs out first

BEFORE THE DAY — FOLLOW

The wiring run, walked once by whoever last did it

A coaxial line goes from the control rack to the mixing chamber through four temperature stages and a dozen clamps, and the route matters because every clamp is a heat leak somebody has to justify. Stay with them and note where it is anchored — that is the part nobody writes down.

PLAN CARD 142W

Monday, and the coherence numbers are lopsided. Anders Holm has the relaxation time on every qubit sitting about where he wants it and the coherence time at a fraction of that, on all 12 within a few per cent — and a qubit's coherence can never exceed twice its relaxation time, so whatever is doing this is scrambling phase without taking any energy at all. Ijeoma Okafor says the fridge is quiet and has a fortnight of logs to show it. Ren Nakamura suspects the flux line. Priya Raghavan needs to know which of these numbers goes in the review and what it means for the circuit she has promised. Today you read both times off the traces yourself, work out what a poor coherence time against a good relaxation time is telling you, and decide where the noise is getting in.

BRIEFING 18W

T1 is respectable and T2 is a third of what it should be, on every qubit at once.

WHAT YOU DECIDE 9W

Separate energy leaving the qubit from phase being scrambled.

3.1 Energy out, or phase scrambled

ERROR & VERIFICATION · A ROOM · SWEEP

SCENE 62W

Two measurements run at every delay. One asks how much excitation is left in the qubit. The other asks how much phase coherence is left. Both start near one and fall as the delay grows. By convention a decay time is read where a quantity has dropped to about a third of where it started, and Holm needs the phase curve's time.

GUIDE 63W

Drag the delay. Both curves are plotted wherever you stop, and each readout gives that curve's value there. You want the delay at which the phase curve, the Ramsey contrast, has fallen to about a third of its starting value. Read it off the plot rather than working it out, then mark that delay and commit. Only the position you mark is graded.

ASK 21W

Both curves fall, and not together. At what delay has the phase coherence dropped to a third of where it started?

BEHIND THE BUTTON 209W

What the two curves are. One is energy: a qubit in the excited state eventually gives its energy away, and the excited population decays as it does. The other is phase. A qubit can keep all of its energy and still lose the phase relationship that makes it useful for computing, because the phase advances at a rate set by its frequency, and anything that shifts that frequency between repetitions scrambles it. The two are separate losses with separate cures.

Why they are read at the same place. A decay is described by the time it takes to fall to about a third of its starting value — one factor of e — and reading both curves at the same fraction is what makes the two times comparable. Read one at a third and the other at a half and the ratio between them means nothing.

Why the ratio is the point. Energy loss also destroys phase, so the phase time can never be much more than twice the energy time. If it is far shorter than that ceiling, something is scrambling phase without taking any energy — and that something is usually noise slow enough to look like a small frequency shift from one repetition to the next.

TAKES AS READ

a quantum state has both an energy and a phase, which can be lost separately · a decay time is conventionally read where a quantity has fallen to about a third of its start

VERDICT 107W

Read at the same place on both curves, the two times come out a factor of three apart, and that gap is the diagnosis. Energy is leaving slowly; something is scrambling phase three times faster without taking any energy at all, which is what slow noise shifting the qubit's frequency between repetitions does. The two have

different cures — energy loss is materials and fridge, dephasing is usually wiring — so the ratio decides who spends the week. It is also a bound worth carrying: coherence can never exceed twice the relaxation time, so this device could have reached 180 and is managing a fifth of that.

TAKEAWAY 17W

Comparing T1 with T2 tells you whether energy loss alone can explain the observed loss of coherence.

3.2 How good T2 could be

CRYOGENICS & VACUUM · A PERSON · BALLPARK

SCENE 35W

Okafor wants to know how much room there is before anybody spends a week chasing the noise. The relation is one over T2 equals one over twice T1 plus one over the pure dephasing time.

GUIDE 71W

Five numbers, two of which are the same relaxation time written with and without its factor of two. The relation is written in rates. A rate is the reciprocal of a time, so the factor has to sit where the algebra put it. Read the equation before reading the labels. Halving or doubling that term by mistake shifts the dephasing time enough to send somebody after noise that is not there.

ASK 8W

Estimate the pure dephasing time behind these numbers.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

two independent loss processes add as rates rather than as times · superposition as an amplitude, not a mixture — taken as read

VERDICT 69W

Under the simple exponential model, $1/T_2 = 1/(2T_1) + 1/T_\phi$. The first term is the phase loss that accompanies energy relaxation; the remainder is labelled pure dephasing. Subtracting rates gives a useful diagnostic number, not a proof of a microscopic mechanism. If the observed decay is strongly non-exponential, as can happen with slow noise, the model itself has to be revisited rather than forcing every trace into one T_ϕ .

TAKEAWAY 18W

When independent exponential decoherence processes are modelled as rates, the rates add and the corresponding times do not.

3.3 What scrambles a phase without taking energy

FABRICATION & MATERIALS · A ROOM · PROTOCOL

SCENE 27W

Lindqvist lists four candidates on the board and asks which of them changes a qubit's frequency slightly and slowly, rather than taking a quantum out of it.

GUIDE 69W

The four sources on the left are all real noise in this device. The four descriptions differ on one thing: whether the process takes energy out of the qubit, or only moves its frequency between repetitions. Ask that question of each source before pairing anything. A mechanism sorted into the wrong channel sends a week of shielding work at the wrong target, and the coherence number does not move.

ASK 6W

Say where slow noise comes from

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a qubit's frequency can depend on its surroundings

VERDICT 88W

Dephasing appears when the qubit accumulates a different phase from run to run. Flux noise and bias-current noise can do that by shifting the transition frequency. Residual photons in a dispersively coupled readout resonator can also shift the qubit frequency randomly and dephase it. A resonant material defect can exchange energy with the qubit and often shows up strongly in T1, although real defects can affect more than 1 coherence channel. The classification here is therefore about the dominant signature in this situation, not a universal one-to-one rule.

TAKEAWAY 23W

A process that changes the qubit frequency from one repetition to the next can wash out phase even if little energy is lost.

3.4 What is worth doing at the wrong temperature

CONTROL & READOUT · A ROOM · CALLBACK · TRIAGE

SCENE 37W

Fixing the heat leak means a warm-up and another cooldown: four days. With Delft's slot approaching, Mensah does not want four empty days and asks what work remains interpretable while the device is off its normal temperature.

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readout and benchmarking should be repeated after the fridge returns to its normal operating point.

TAKEAWAY 18W

An off-nominal machine can still produce useful diagnostics, but those data must not be mistaken for nominal performance.

END OF THE DAY 18W

Relaxation and dephasing remove different parts of quantum information, so the ratio of T1 to T2 is diagnostic.

One bit out

PLAN CARD 102W

Tuesday, and the argument is about a single number. The single-shot readout fidelity is quoted at 96%, and Ren Nakamura points out that it is the product of 3 separate things — how well the resonator separates the 2 states, how much noise the amplifier chain adds, and where somebody drew the line between them. Anders Holm wants each of the three measured. Priya Raghavan wants the headline figure for the review. Today you work out what the readout chain does to a signal, what a discriminator is actually deciding, and what part of that 96% belongs to the qubit at all.

BRIEFING 15W

The readout says 96% fidelity. Nakamura says that number is doing a lot of work.

WHAT YOU DECIDE 14W

Understand what a readout actually decides, and how confident it is entitled to be.

4.1 What happens between the chip and the answer

CONTROL & READOUT · A ROOM · SEQUENCE

SCENE 39W

The device is cold, the qubit is located, and its coherence is measured. Before Ridgeway sends the experiment to Delft, Nakamura walks the final readout path: resonator, cable, isolator, parametric amplifier, 4-kelvin amplifier, more cable, then the room-temperature digitiser.

GUIDE 77W

The four cards are places in a chain rather than moments on a clock, so nothing here happens before anything else in time. What orders them is where the signal is: how cold, how far from the chip, and how much gain it has already been given. Work outward from the qubit and let the temperature be the guide. Put the gain stages the wrong way round and the chain's noise is set by its warmest amplifier.

ASK 4W

Follow the signal out

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

an amplifier increases a signal and adds noise of its own

VERDICT 92W

"A useful rule comes from the noise budget for cascaded amplifiers: once the signal has been given large low-noise gain, the noise of later stages is divided by that gain when referred back to the input. That makes the first high-gain amplifier extremely important. Loss before it is just as serious because it throws away signal before the gain arrives. The cold parametric amplifier is therefore placed close to the device and behind only the components needed to protect and route the signal; the warmer semiconductor amplifier then supplies robust additional gain."

TAKEAWAY 20W

In a cascaded readout chain, noise added early — and loss before the first high-gain amplifier — is especially costly.

4.2 A line drawn between two clouds

CONTROL & READOUT · A PERSON · SWEEP

SCENE 60W

The digitiser's shots plot as two clouds, ground and excited. Most sit clearly inside one of them. A few thousand lie in the gap between, and a threshold anywhere in that gap assigns every one of those to a state. Move it one way and ground shots get called excited. Move it the other and excited shots get called ground.

GUIDE 62W

Drag the threshold. Two curves are plotted: ground shots called excited, and excited shots called ground. Both mistakes cost the same in this circuit, so what matters is the two added together rather than either one alone. Sweep the range and watch both readouts, then mark the threshold you would set and commit. You are graded on where you put the line.

ASK 22W

Both mistakes cost the same in this circuit. Where should the line go, and what does the total misassignment there tell you?

BEHIND THE BUTTON 161W

Why the clouds overlap. A readout is a voltage built up over a few microseconds, and it carries amplifier noise. Some shots also change state during the measurement itself, which puts them between the two clouds by right rather than by accident. So the overlap is physics, not sloppiness, and no threshold can remove it.

How the two curves behave. Each one falls as the other rises: every millivolt you move the line rescues some shots of one kind and loses some of the other. The two therefore have a total with a shape, and it is that total the circuit pays.

Why this line matters more than it looks. Every number built from classified outcomes inherits this threshold — readout fidelity, circuit fidelity, the correlations later in the fortnight. A threshold set in the wrong place does not announce itself; it quietly lowers whatever is reported downstream, which is a thing that happens to this group again on day 10.

TAKES AS READ

a measurement produces a distribution rather than a single value

VERDICT 122W

The line does not remove ambiguous shots; it decides which state they are called, so every millivolt taken off one error is added to the other. That is why the two curves cross and why the sum has a minimum that is not zero. The floor is the overlap between the two clouds, and it is a property of the measurement rather than of the threshold — getting under it needs more signal, less noise or a longer integration, something that moves the clouds apart. Equal costs put the line at the crossing; where one mistake costs more than the other, the best line moves away from it, which makes this a question about the circuit and not only about the classifier.

TAKEAWAY 20W

A discriminator converts overlapping analogue distributions into reported states, so its threshold sets a trade between two kinds of misclassification.

4.3 Whose error is it

ERROR & VERIFICATION · A ROOM · CASEBOOK

SCENE 37W

The first end-to-end diagnostic is in. It reports 8% wrong outcomes. Separate calibration experiments estimate about 4% readout error and less than 1% preparation error. Holm asks what Ridgeway can honestly infer about the remaining circuit contribution.

GUIDE 74W

The left column is four terms of one error budget; the right says what kind of evidence each one is. They differ in how each was got: measured on its own, seen only inside a total, or inferred once the others are modelled. Ask of each which of those three it is. A budget that quotes an inferred term as a measurement sends a pulse engineer to optimise what the readout chain is doing.

ASK 8W

Take the readout out of the qubit number

BEHIND THE BUTTON 181W

Why explanations rather than labels. Naming a finding is not accounting for it, and a plausible-sounding mechanism attached to the wrong observation is the commonest way a wrong story survives. Committing an explanation to one clue means claiming it accounts for that clue specifically and not for its neighbour, which is where the two come apart.

How to use the one-each rule. The explanations are a set to be distributed, not a list to be sampled, so every join constrains the rest. Settling the two you are confident of can decide the remaining pair by elimination. Where it does not, two explanations are still competing for one clue, and that competition is the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every explanation used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

errors from independent sources combine

VERDICT 79W

"For small, approximately independent error probabilities, adding the contributions is a useful first-order budget, but it is not an exact law. The 8% observed error therefore suggests that several per cent remain for gate and circuit effects after accounting for readout and preparation. A more careful analysis would model how the errors compose rather than simply subtracting percentages. The educational point is the separation: the number a user sees is not automatically the number a pulse engineer should optimise."

TAKEAWAY 20W

An observed error rate is only a diagnosis after you state the model used to separate preparation, gates and measurement.

4.4 Which stage the leak is on

CRYOGENICS & VACUUM · A ROOM · CALLBACK · PROBE

SCENE 61W

A dilution refrigerator cools in stages. Each one is colder than the stage above it: 50 kelvin, 4 kelvin, the still, the heat exchanger, the cold plate, and the mixing chamber at the bottom. Heat travels down that chain and stops where it is anchored.

Ridgeway's fridge has stalled warm. Okafor has last cooldown's numbers and a thermometer on every stage.

GUIDE 75W

Nothing has been measured for you. Press Read on any stage and it reports what it holds now, what it held last cooldown, and how hard its cooling is working. Take at least three, and remember that a stage matching last time tells you as much as one that does not. Then click the stage where the heat is arriving and commit. You are graded on the stage, not on how many readings it took.

ASK 28W

The bottom of the fridge is warm and the top may not be. Take the readings you think you need — at which stage is the heat arriving?

BEHIND THE BUTTON 196W

How the stages work. Each stage is anchored to the one above by a deliberate thermal link, and cools whatever is bolted to it. A stage removes a fixed amount of heat per second, so extra heat arriving does not simply pass through: the stage warms and its cooling works harder, and everything below it is left with less cooling to spare. That is why a load shows up as a pattern down the chain rather than at one thermometer.

What the numbers mean. "Holding now" against "last cooldown" is a comparison with a known-good run, which is worth more here than any absolute temperature — a stage sitting at its usual value under its usual load has been ruled out, and one sitting high has not. The cooling effort is the third column: a stage can hold its temperature and still be telling you something, if it is working harder to do it.

Why the top of the chain matters. A vacuum leak, a failing pulse tube, or a room-temperature fault all deliver heat to the warmest stages first. If those are unchanged, a whole family of explanations is gone before you look further down.

TAKES AS READ

a refrigerator removes heat in stages, each colder than the last · a conductive path carries heat to wherever it is anchored, and past wherever it is not · thermal energy against level spacing — why a qubit needs millikelvin — taken as read

VERDICT 158W

Two stages held exactly what they held last time, which is what rules almost everything out: a vacuum problem, a failing pulse tube, a room-temperature fault would all warm the top of the chain first, and the top of the chain has not moved. From the still downwards every stage is high and working harder, and heat only

behaves that way when it is being delivered from above. So the deepest stage is the worst affected and is not where the problem is — the load enters at the first stage that departs, and the stages below it are inheriting it. That is a conductive path anchored at 50 K and 4 K and not below, which is exactly what a line added since the last cooldown and heat-sunk at only the easy stages does. Okafor's cable is now worth looking at, not because cables are always guilty but because this pattern is what an unanchored one predicts.

TAKEAWAY 17W

A refrigerator's stage-by-stage temperature pattern can localise a heat load instead of merely announcing that one exists.

END OF THE DAY 16W

A readout fidelity belongs to the entire measurement chain: qubit-state-dependent signal, loss, amplification, noise and classification.

The offer with a date on it

PLAN CARD 127W

Wednesday, and a note arrives from a group in Delft. They have a device of a different design and a window of machine time in ten days, and they will run Ridgeway's circuit on it if they get the logical gate sequence, preparation and measurement protocol by Friday. Anders Holm's instinct is to send nothing until the local numbers are settled, because a protocol sent out creates work and expectations the group no longer controls. Priya Raghavan points out that this is the second device he has been asking for since March, offered free, with a deadline. Today you decide what has to be shared for a genuinely independent check, what should stay hidden until after the run, and what the group gives up if it waits.

BRIEFING 23W

A group in Delft will run your circuit on their device this month, if they get the gate sequence and protocol by Friday.

WHAT YOU DECIDE 14W

Decide what to share, and when, while somebody else is deciding the same thing.

5.1 What you would need back

ERROR & VERIFICATION · A ROOM · PROTOCOL

SCENE 43W

An email from Delft is open on Holm's screen: their independent-device slot begins in ten days, and the next opening is a quarter away. Holm starts a list of exactly what Ridgeway would have to send, and what would have to come back.

GUIDE 63W

The four items differ in one respect. Does sharing each one make the second run a repeat of the experiment, or a repeat of Ridgeway's own assumptions? Ask whether Delft could reach the same conclusion without it. A replication that shares the analysis as well as the method can only agree. It has given up the independence that made a disagreement worth having.

ASK 8W

Decide what makes a replication worth the name

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a second measurement is only a check if it could disagree · calibration drift, and the tuning that has to happen daily — taken as read

VERDICT 79W

An independent run should reproduce the scientific procedure without needlessly reproducing Ridgeway's hidden assumptions. Delft needs the circuit and enough implementation detail to run the same intended experiment. Ridgeway should ask for the raw single-shot results so both groups can examine them, but Delft should build its own readout rule and device calibration. The expected numerical result should also be withheld until Delft has frozen its analysis. Shared instructions define

the experiment; independent calibration and analysis make disagreement informative.

TAKEAWAY 20W

A useful replication shares enough method to repeat the experiment while keeping independent the choices that are actually being tested.

5.2 Copying a state is not the same as transferring one

NETWORKS & SECURITY · A PERSON · CHOICE

SCENE 39W

Yusuf Sadiq points out that Delft does not need Ridgeway's physical quantum state at all; it needs the recipe for preparing its own. An arbitrary unknown state cannot be turned into a classical file that produces two perfect copies.

GUIDE 67W

All four options agree that something cannot be sent, and disagree about what. The candidates are the size of a description, the transfer itself, and what copying a recipe does. Separate the classical object here, the preparation instructions, from the physical object, the prepared state. Then ask which of the two each option is really about. Whether quantum networks are possible at all turns on that distinction.

ASK 24W

Why can Ridgeway email the preparation instructions but not email a complete classical description that lets Delft reconstruct an arbitrary unknown prepared state exactly?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

classical descriptions can be copied freely, while an arbitrary unknown quantum state cannot be perfectly cloned

VERDICT 87W

A classical preparation recipe is ordinary information and can be copied without limit. An arbitrary unknown quantum state is different: there is no universal physical operation that takes one copy and produces two perfect copies of it. That is the no-cloning theorem. It does not say quantum information can never travel; a physical quantum system can be transferred. The restriction is that the original

cannot remain behind as a second perfect copy. For this replication Delft needs the preparation instructions, not Ridgeway's particular instance of the state.

TAKEAWAY 21W

The no-cloning theorem forbids making a perfect duplicate of an arbitrary unknown quantum state; it does not forbid transferring quantum information.

5.3 What goes out on Friday

QUANTUM SENSING · A ROOM · SCIENCETANK

SCENE 44W

Elena Barros, who runs Ridgeway's sensing group and ships calibrated numbers to a hospital every month, has the least romantic view of the Delft offer: if somebody else will depend on Ridgeway's work, the method has to leave the building before the opportunity does.

GUIDE 88W

Delft's window begins in ten days and the next is a quarter away. Their device is a different design, which is the whole reason a second run counts as independent. Ridgeway's result has been measured on one device by one group. Delft has asked for sequences, not for results. Knowing the expected number before freezing an analysis changes every choice made along the way. The external review follows soon after the run. A result published first is one the second run is compared against rather than tested against.

ASK 6W

Decide what goes out on Friday

BEHIND THE BUTTON 193W

Why the whole spread is graded. Funding is not a vote for one idea. A portfolio says what you think is likely, what is worth hedging against, and what is not worth pursuing at all — and the last two are where most of the information is. Backing the right proposal while quietly funding a bad one is a worse answer than it looks. That is why the small numbers count as much as the big one.

What the three numbers are for. Thirty-five is what makes a lead a lead: below it you have hedged rather than chosen. Fifteen is the most that can sit on unsupported work before it stops being a rounding error. Past that it is a second opinion nobody argued for. Twenty is the floor under a line of work you have already called strong, because funding it too thin to finish spends the money and buys nothing.

Why there is a floor on the total. Points held back are not caution; they are a decision not to decide, taken with somebody else's money and somebody else's deadline. The floor is what forces the panel to say something.

TAKES AS READ

a result can be shared at different levels of completeness · calibration drift, and the tuning that has to happen daily — taken as read

VERDICT 81W

The Delft window is short and the point of using a different device is to create an independent test. Sending the circuit, implementation notes and stated uncertainties does not require Ridgeway to send its expected result or claim that its own number is final. Waiting until every local number is polished forfeits the independent run. Sending the expected value with the method would weaken the check, while

publishing before the cross-check turns an opportunity for pre-publication diagnosis into a post-publication dispute.

TAKEAWAY 19W

Sharing a reproducible method before the final claim is settled can create evidence without committing to the final answer.

5.4 The energy of one quantum against the energy of the room

CRYOGENICS & VACUUM · A ROOM · CALLBACK · BALLPARK

SCENE 38W

The chip's qubits answer at about 5 gigahertz. Okafor circles the Delft date on the whiteboard and writes two energy scales beside it: one quantum at that frequency, and the thermal energy at 42 millikelvin. Then she waits.

GUIDE 75W

Five numbers, and three of them are energies while two are the raw frequency and temperature those energies were built from. Two of the energies are thermal, taken at different temperatures. The exponent is a ratio of energies, so both terms have to be in joules and both have to describe today's fridge. Using the colder thermal energy answers a question about a machine that is working. That is not the one on the bench.

ASK 13W

Using the 2 energy scales, estimate the equilibrium excited-state fraction at 42 mK.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a system in contact with something warm can absorb energy from it

VERDICT 91W

"A qubit sitting in a warm environment does not stay in its ground state: it is being offered energy all the time, and it takes some. The fraction that ends up excited follows the ratio of the thermal energy to the energy of one quantum, exponentially, which is why a factor of four in temperature is not a factor of four in behaviour. At 11 millikelvin almost nothing is excited. At 42 the population is large enough that a measurement of \"the ground state\" is partly a measurement of the fridge."

TAKEAWAY 17W

Thermal energy well below one quantum is what keeps a qubit in its ground state at rest.

END OF THE DAY 21W

Some evidence has a window on it, and refusing to move inside that window is a decision with its own cost.

A defect with a date on it

BEFORE THE DAY — HUNT

Six attenuators fitted, and the diagram shows four

Every line into the fridge is attenuated at each stage, and the diagram has not been updated since somebody added two more during the last cooldown. Until all six are found nobody can say what power actually reaches the chip, and every calibration this week is a number divided by an attenuation nobody has confirmed.

PLAN CARD 112W

Thursday, and Sara Lindqvist has been through the drawer. The weak absorption sitting 60 megahertz below the twelfth qubit appears on 3 chips, and all 3 came from wafers processed in a six-week window last spring — the same window in which the oxidation step was changed to raise yield. Kofi Mensah has watched a qubit lose energy every time its frequency drifts into that feature. Priya Raghavan has to decide whether this goes in the review as a known cause or as an open question. Today you work out what the feature is, what it does to a qubit that sits near it, and what the process change bought and cost.

BRIEFING 17W

The weak response near 4.55 GHz turns up on 3 chips, all from the same 6 weeks.

WHAT YOU DECIDE 13W

Trace a fault in a device back to the process that made it.

6.1 Something else that absorbs at that frequency

FABRICATION & MATERIALS · A ROOM · DIAGNOSIS

SCENE 42W

By breakfast Lindqvist has three chips from one processing window lined up on the bench. The same narrow feature appears on all three and shifts after thermal cycling. Chips from every other window are clean. The fabrication log is open beside them.

GUIDE 71W

Five readings are on the panel, and the unremarkable ones carry as much weight as the narrow line. Three chips from one process window, clean chips from every other, and a shift after cycling: each rules out a different candidate. Test every explanation against all five readings, not the loudest. An explanation that fits the feature but not where it appears is a hypothesis about the wrong part of the building.

ASK 6W

Which explanation fits all of it?

BEHIND THE BUTTON 179W

Why the unremarkable readings decide it. The salient reading is what draws attention, and it is usually consistent with several explanations at once, which is why it rarely settles anything. The readings that discriminate are the ones a candidate predicts should have moved and which have not: a normal value is a positive result against every mechanism that would have disturbed it.

How to work the candidates. Take each mechanism and predict the panel it implies before you look at the panel again — which readings it drives, in which direction, and by roughly how much. Then compare. Working that way round is what separates a diagnosis from a rationalisation, because the prediction is made before the data is consulted.

Why only one candidate survives. Several will account for part of the panel, deliberately so, and a partial fit is exactly what a confident wrong answer feels like from the inside. When two remain, look for the reading on which their predictions differ and let it decide. If no reading separates them you have not finished reading the panel.

TAKES AS READ

a material can contain flaws that behave as small two-state systems

VERDICT 87W

The evidence strongly favours a device-local material resonance. The feature appears only on chips from one process window, survives changes in shared room-temperature electronics, and shifts slightly after thermal cycling. A neighbouring designed qubit or a generator artefact should have different cross-chip behaviour, while thermal photons usually change linewidths and backgrounds rather than producing the same isolated chip-specific line. A two-level material defect is therefore the leading hypothesis. Calling it specifically a defect in the junction oxide would require more localisation evidence than this spectrum alone provides.

TAKEAWAY 23W

A narrow, repeatable spectral feature can be used as a fingerprint, but identifying its microscopic cause requires ruling out competing modes and artefacts.

6.2 Energy with somewhere to go

CONTROL & READOUT · A PERSON · SWEEP

SCENE 46W

This qubit can be tuned across about 400 megahertz, and Nakamura has measured how long it keeps its energy at every setting in that band. That relaxation time is the readout, and longer is better. Somewhere inside the band sits the defect the group found yesterday.

GUIDE 67W

Drag the qubit frequency. The readout is the relaxation time in microseconds, and higher is better. Across most of the band it is roughly flat, and you are looking for the narrow place where it collapses, because that is the frequency to keep this qubit away from. The damage is only tens of megahertz wide, so sweep the whole band before deciding. Then mark it and commit.

ASK 25W

Relaxation is around 90 microseconds across most of the band and much worse in 1 place. Which frequency should this qubit be kept away from?

BEHIND THE BUTTON 151W

Why a frequency can cost you energy. Two systems at the same frequency exchange energy easily; at different frequencies they barely talk. A defect in the oxide with a transition of its own is therefore harmless almost everywhere and ruinous where the qubit's frequency matches it, because the qubit's energy has somewhere to go.

Why the dip is narrow. The width of the damage is the defect's own linewidth, which is why it is tens of megahertz rather than hundreds, and why a qubit parked 50 megahertz away is unaffected. It also means a coarse sweep can step straight over it.

What this turns a measurement into. A loss channel that is sharply resonant makes frequency an engineering control rather than merely a property: the fix is not to repair the defect but to operate away from it. That is why a tunable qubit is worth the extra circuitry it costs.

TAKES AS READ

two systems at the same frequency can exchange energy

VERDICT 115W

A qubit loses energy quickly when there is somewhere for that energy to go, and a two-level defect is somewhere — but only while the two are close in frequency, which is why the loss appears and disappears over tens of megahertz rather than across the whole band. Two things follow. The width of the dip is a measurement of how strongly the two are coupled, and on a tunable device the frequency itself becomes an engineering control: move the qubit and the loss channel is not there at all. The half of the job that gets forgotten is writing the band down, because an undocumented defect is a trap for whoever retunes next month.

TAKEAWAY 19W

If a loss channel is sharply resonant, frequency can be an engineering control knob as well as a measurement.

6.3 Yield against lifetime

ERROR & VERIFICATION · A ROOM · TRIAGE

SCENE 24W

The oxidation change raised the fraction of usable chips from about half to nearly three quarters. It also appears to have produced these defects.

GUIDE 71W

All four options are defensible, and they differ in what each treats as already established. One correlation across three chips from a single window is a reason to run a test, not a finding. Ask of each option what evidence it would need before it became the right move. Trading a measured yield gain for an unmeasured coherence gain is the kind of decision that looks careful and costs a quarter.

ASK 9W

What should the group do about the oxidation step?

BEHIND THE BUTTON 194W

Why first is a different question from most important. When several calls compete for the same hour, what a choice is worth is not its own importance but the difference between doing it now and doing it later. A serious problem that will be no worse in an hour costs nothing to defer. A smaller one that closes a door costs everything behind that door.

What to look for in the options. Two things separate them: which is still changing, and which is a precondition for the others. A situation that is deteriorating has a cost per hour attached to it, and a step that unblocks the rest multiplies the value of the hours after it. Everything else is a preference about where to start.

Why only one answer is marked. In the situation these options describe all of them eventually happen; what is being tested is the head of the queue, because that is where the reasoning is visible. The verdict names what each of the others was waiting on, which is worth reading even when the choice was right — the ordering behind the first place is the rest of the answer.

TAKES AS READ

a process change can improve one measure and damage another

VERDICT 80W

Three affected chips from one processing window are enough to motivate a test and not enough to identify the oxidation change as the cause. The process change also produced a measured yield gain, so reverting immediately trades a known benefit for an unmeasured one. The clean next experiment is to fabricate comparable material under the old and new conditions, then measure both yield and defect density with the same protocol. That turns a suggestive time correlation into a causal test.

TAKEAWAY 21W

A process change can improve one metric while degrading another, so causation needs a controlled comparison rather than a retrospective correlation.

6.4 Ten millikelvin, while the wiring warms

CRYOGENICS & VACUUM · A ROOM · CALLBACK · HOLD

SCENE 38W

The chip has to sit at ten millikelvin while a measurement runs. A new coaxial line is carrying heat down from room temperature, the still is working, and the flow control is the only thing that answers it.

GUIDE 49W

Hold the mixing chamber temperature inside the band on the thermometry. The band narrows through the run, because the same excess temperature costs more the longer the qubit has to stay coherent. The flow control is your control, and every heat load keeps arriving until the circulation answers it.

ASK 11W

Hold the mixing chamber at ten millikelvin while the measurement runs.

BEHIND THE BUTTON 129W

Why ten millikelvin and not one kelvin. A qubit's two levels are a few gigahertz apart, and thermal energy at one kelvin is comparable to that spacing — so the qubit sits in a mixture rather than in its ground state, and every experiment starts from noise.

Why a heat load is a rate. A coaxial line conducting from room temperature does not raise the chamber by a millikelvin and stop; it carries heat continuously, so the temperature climbs until the circulation carries the same power away.

Why the margin tightens. The longer a measurement runs, the more of the qubit's coherence has already been spent, so a warm spell late in a run costs a larger share of what was left than the same excursion at the start.

TAKES AS READ

a system in thermal contact settles toward the temperature around it

VERDICT 161W

A superconducting qubit needs millikelvin because its two levels are only a few gigahertz apart, and thermal energy at a kelvin is comparable to that spacing — the qubit would sit in a thermal mixture rather than in its ground state, and every shot would start from noise. Holding that temperature is a continuous job because every disturbance is a rate. A coaxial line conducts heat down from room temperature for as long as it is installed; a pulse tube cycling changes the load on the still continuously; a helium level falling changes the cooling power available. None of them is a step, so the flow is set to a new rate and left there rather than nudged. The band narrows for the same reason the whole experiment is timed: coherence is being spent as the run goes on, and a warm spell in the last minutes costs a larger share of what remains than the same excursion in the first.

TAKEAWAY 13W

A heat load that keeps arriving is answered by circulation, not by waiting.

END OF THE DAY 26W

A defect in a material shows up months later in somebody else's measurement, which is why the record of what was made when is worth keeping.

The instrument somebody else depends on

PLAN CARD 109W

Friday, and Elena Barros is not interested in the review. Her magnetometer measures fields 100 million times weaker than the Earth's, a hospital across the city uses those numbers in a study that has been running for 3 years, and the calibration is due at 2 o'clock whatever else is happening. Grace Whitfield is bringing the reference. Priya Raghavan, who needs Barros for the review slides, has been told to come back at three. Today you work out where a sensor's precision limit comes from, what averaging can and cannot buy, and why a number that leaves the building carries obligations that a number in a slide does not.

BRIEFING 15W

The magnetometer has to be recalibrated by two, because a hospital is using its numbers.

WHAT YOU DECIDE 16W

Meet the part of this field that already works, and the discipline that comes with it.

7.1 What averaging buys

QUANTUM SENSING · A ROOM · SWEEP

SCENE 50W

One measurement takes 100 milliseconds and resolves 40 picotesla. Averaging independent measurements reduces random error, so averaging more of them resolves a smaller field. Here the smaller number is the better one. Barros has an hour of instrument time, and wants to know how much of it is worth spending.

GUIDE 63W

Drag the integration time. The readout is the field the instrument can still resolve, in picotesla, and lower is better. Averaging helps only while the measurements stay independent, and the curve will show you where that stops being true. Mark the integration time you would actually use, then commit. You are graded on the time you choose, not on the resolution you reach.

ASK 20W

Averaging for longer buys resolution, until it does not. How long should she average, and what stops her going further?

BEHIND THE BUTTON 175W

Why averaging works, and what it costs. Random noise on N independent measurements averages down as the square root of N , so the resolution improves as $1/\sqrt{N}$. That is a harsh exchange rate: the first factor of ten in resolution costs a hundred measurements, and the next factor of ten costs ten thousand. Sensing is a patience problem rather than a precision problem, and the arithmetic is why.

What breaks the independence. The formula assumes each measurement's noise is unrelated to the last one's. Real instruments drift — temperature, bias, the field of the room itself — and a drift is shared by every measurement taken during it. Once the drift is bigger than the averaged noise, more averaging makes the number worse rather than better, because you are averaging in something that is not noise.

What the readout means. The resolvable field is the size of a change this instrument could still see: roughly one standard deviation of the averaged noise. Reporting a field smaller than that is reporting the noise.

TAKES AS READ

independent measurements can be averaged to reduce random error

VERDICT 112W

Random noise averages down as the square root of the number of independent measurements, so the first factor of 10 in resolution is cheap and the next costs 100 times the time — the reason sensing is a patience problem rather than a precision 3 things then fight. Averaging improves the number only while the measurements are independent, and past about half a minute the instrument itself has drifted, so later measurements are of a slightly different machine. Beyond the knee the curve turns back up and more data makes the answer worse. That is also why the hour does not help: the limit is drift, not the number of shots available.

TAKEAWAY 16W

For independent measurements, random uncertainty falls only as $1/\sqrt{N}$, so time buys precision with diminishing returns.

7.2 Against what

QUANTUM SENSING · A PERSON · CHOICE

SCENE 24W

Whitfield arrives with the reference source at two. Barros measures it, writes the ratio in the log, and only then measures anything anybody wants.

GUIDE 73W

Every option here describes something a measurement can genuinely establish. They differ in whether the subject is scatter, scale, or the sample itself. Repetition tells you how much a number moves; it cannot tell you whether the whole scale is displaced. Decide which of those two a reference is for. In a study running over years, an untested scale lets a drift in the instrument arrive looking like a trend in the patients.

ASK 16W

What does today's comparison with the reference establish that repeated measurements of the hospital sample cannot?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

an instrument's reading is a comparison · resonance, detuning, and finding a qubit frequency — taken as read

VERDICT 66W

Repeated measurements can tell Barros about precision and short-term stability, but they cannot reveal a scale error that repeats every time. Comparing with a reference of known value tests the instrument's scale and creates a traceable link between today's reading and previous ones. For a multi-year study, that link matters because an uncorrected drift in the instrument can masquerade as a trend in the patient data.

TAKEAWAY 16W

Precision describes scatter; calibration against a reference is what connects that scatter to an accurate scale.

7.3 Numbers that leave the building

NETWORKS & SECURITY · A ROOM · PROTOCOL

SCENE 31W

Sadiq points out that his key link has the same property: somebody else's system depends on it, so its failures have to be visible to them rather than only to him.

GUIDE 63W

The left column is four things a delivered measurement has to carry with it. The right column is four ways a user gets hurt. Pair them by asking which failure each obligation makes impossible, not which one sounds most responsible. None of these makes the instrument more sensitive. They decide whether somebody outside the building can tell when the sensitivity has stopped applying.

ASK 8W

Say what a delivered number obliges you to

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a service somebody relies on has different obligations from an experiment ·
superposition as an amplitude, not a mixture — taken as read

VERDICT 82W

A research measurement can be repeated before anybody acts on it. A delivered measurement may already have changed a decision, so the user needs enough context to recognize when the service is outside its validated state. An uncertainty supports decisions about significance; a calibration interval supports comparisons through time; a visible fault flag prevents silent use of known-bad data; and a change log makes later re-analysis possible. None of these substitutes for better sensitivity, but they determine whether the sensitivity is usable.

TAKEAWAY 20W

Once another person acts on a measurement, uncertainty, alarms and change history become part of the information you are delivering.

7.4 Where the qubit actually is

CONTROL & READOUT · A ROOM · CALLBACK · SWEEP

SCENE 57W

A qubit absorbs a drive only near its own transition frequency. So you find that frequency by sweeping the drive and watching how much of the population ends up excited. The design put this one at 5.02 gigahertz. The readout there has been flat all morning, so the device that came out of fabrication is somewhere else.

GUIDE 69W

Drag the drive frequency. The readout is the fraction of shots that finish excited, so a frequency this qubit absorbs shows up as a rise, and only the places you stop at get plotted. Sweep the whole band before deciding: more than one feature can appear, and they are not the same size. What counts as the transition is on the Background button. Then mark the frequency and commit.

ASK 15W

Nothing responds at the design frequency. Which frequency does this qubit absorb the drive at?

BEHIND THE BUTTON 170W

Why the design number is only a guess. The transition frequency is set by a junction barrier about a nanometre thick, and no fabrication process controls a nanometre exactly. A device landing a few hundred megahertz from its design value is ordinary, which is why every qubit is found by spectroscopy before it is used.

What a resonant drive does. On its own transition, a drive moves nearly the whole population into the excited state, because every repetition is driving the same transition in step. Off resonance the population barely moves. So the size of a feature is evidence about what it is, not merely about how strong the drive was.

What else can absorb. A smaller feature elsewhere in the band is usually real and usually not the qubit: a two-level defect in the oxide, a stray mode of the package, or a neighbouring qubit. Those are worth knowing about — one of them turns up again later this fortnight as a place this qubit must not be parked.

TAKES AS READ

a system absorbs energy strongly only near its own transition frequency

VERDICT 108W

A qubit absorbs a drive only close to its own transition, so the frequency where the population goes up is the transition. It is nowhere near 5.02 GHz because that number was a prediction: the frequency depends on a junction barrier about a nanometre thick, and no process controls a nanometre exactly. Two responses appear, and size is what separates them — a transition driven on resonance takes almost the whole population, while the small feature takes a sixth of it and is something else on the chip. Which of the two is the qubit can be settled by driving harder and watching only one of them saturate.

TAKEAWAY 16W

A designed transition frequency is a target; spectroscopy tells you where the fabricated device actually responds.

END OF THE DAY 19W

Quantum sensing becomes useful outside the lab only when sensitivity, calibration, drift and uncertainty are treated as one system.

How wrong is it, and how would you know

PLAN CARD 102W

Monday, and there are two error figures on the board. A direct measurement of a single gate says 0.4%. Randomised benchmarking over long sequences says 1.2. Anders Holm says both are correct and they answer different questions; Priya Raghavan says the review will only print 1 of them. Dasha Petrova wants whichever number is comparable with the threshold, because below that line adding qubits helps and above it they do not. Today you work out what randomised benchmarking actually measures, why a long sequence gives a different answer from a short one, and which number belongs in a sentence about error correction.

BRIEFING 11W

Two error numbers, both honest, differing by a factor of three.

WHAT YOU DECIDE 14W

Measure the machine's error in a way that does not measure your own readout.

8.1 Why a long sequence tells you more

ERROR & VERIFICATION · A ROOM · SWEEP

SCENE 58W

Every run applies a sequence of gates and finishes with one that should return the qubit to where it started. So each run either comes back or does not. Errors accumulate along the way: the longer the sequence, the smaller the fraction that returns. Holm can run any length from a single gate up to a few hundred.

GUIDE 66W

Drag the sequence length. The readout is the fraction of runs that came back, and only the lengths you stop at get plotted. You want the length at which half the runs fail to return — where the curve crosses one half. Read it off the plot, mark it and commit. The error per gate follows from that length afterwards, and is not what you report.

ASK 15W

How many gates can this qubit take before half the runs fail to come back?

BEHIND THE BUTTON 192W

Why a length rather than a single measurement. If the survival per gate is p , then the fraction returning after n gates is p multiplied by itself n times. A readout error adds a constant offset to every point on that curve — it shifts the whole thing down — but it does not change how fast the curve falls. So a number read from the trend survives an offset that would swamp any single reading, which is the whole reason this experiment is run at many lengths instead of one.

Why the sequence ends by inverting itself. The final gate is chosen to undo everything before it, so a perfect qubit would always come back and the only thing being measured is the error. It also means the answer is the same whatever the sequence was, which is what lets the runs be averaged together.

What this is called and where it goes. Sequences of random gates that invert at the end are the standard benchmark for gate quality, because the number they produce can be compared between laboratories that share no hardware. It is the number a customer asks for.

TAKES AS READ

errors accumulate as operations are applied in succession

VERDICT 126W

Every gate in the sequence has to work for the qubit to come back, so the probabilities multiply and the curve falls exponentially rather than in a straight line. That makes the half-way length a per-gate error measurement: the length itself says how much each gate costs, and 57 gates to lose half of them works out at a bit over 1% each. It is also why the shape matters more than any single point. A readout that mislabels a fixed fraction of shots pulls the whole curve down without changing how fast it falls, so a number read from the decay survives an offset that would ruin a

number read from one sequence — which is the difference between the two figures on the board.

TAKEAWAY 16W

A quantity read from a trend survives a constant offset that would swamp a single reading.

8.2 Where a good number stops being good

CONTROL & READOUT · A ROOM · BALLPARK

SCENE 16W

Mensah's proposed circuit is 400 gates deep. The measured fidelity of a single gate is 0.988.

GUIDE 70W

Five numbers, two of which are the same gate quality written once as a fidelity and once as an error. Another is a benchmark figure in per cent. The relationship is written for one of those forms, so establish which before choosing anything. Raising an error rate to a power where a fidelity belongs gives an answer near zero. It still looks like a probability, and nothing downstream will object.

ASK 19W

Under the simplifying assumption that each of the 400 gates succeeds independently with probability 0.988, estimate the all-gates-success probability.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

independent errors compound over repeated operations

VERDICT 90W

If — and this is a simplifying model — every gate succeeds independently with the same probability and the circuit counts as correct only when every gate succeeds, the all-success probability is 0.988^{400} , about 0.008. The point is the exponential sensitivity to depth. Real quantum errors can be coherent, correlated and partly cancel or amplify, so this product is not a universal prediction of circuit fidelity. It is a useful first estimate that shows why a per-gate number that looks close to one can still be inadequate for deep circuits.

TAKEAWAY 13W

In an independent-error toy model, small per-gate infidelity compounds rapidly with circuit depth.

8.3 The number error correction cares about

ERROR & VERIFICATION · A PERSON · CASEBOOK

SCENE 50W

Petrova wants to write one sentence about error correction. For the simplified stochastic error model used in her code study, the quoted threshold is about 1% in a per-gate error metric intended to match the benchmark. The direct single-gate number and the randomized-benchmarking number sit on opposite sides of it.

GUIDE 78W

Four numbers, all real, all measured on this device, and each produced by a different procedure. A threshold is defined against a particular error metric, so the question is which of these was measured the way that threshold assumes. Match by procedure, not by size or by which side of one per cent a number falls on. Comparing two figures that share a unit but not a definition is how a machine gets declared above threshold on paper.

ASK 7W

Say which number belongs beside the threshold

BEHIND THE BUTTON 181W

Why explanations rather than labels. Naming a finding is not accounting for it, and a plausible-sounding mechanism attached to the wrong observation is the commonest way a wrong story survives. Committing an explanation to one clue means claiming it accounts for that clue specifically and not for its neighbour, which is where the two come apart.

How to use the one-each rule. The explanations are a set to be distributed, not a list to be sampled, so every join constrains the rest. Settling the two you are confident of can decide the remaining pair by elimination. Where it does not, two explanations are still competing for one clue, and that competition is the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every explanation used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a threshold is a comparison against a defined quantity

VERDICT 96W

"For *this stated simplified model*, the 1.2% randomized-benchmarking error is the intended comparison and it lies above the 1% line. The 0.4% direct figure is a best-case measurement of 1 gate under a different procedure, so using it would answer a different question. In real fault-tolerance work, even an average benchmarking error may not be enough: coherent and correlated errors can make the worst-case behaviour relevant to a code much worse than the average infidelity. The careful

sentence therefore names both the metric and the model instead of declaring that the machine is simply 'above threshold'."

TAKEAWAY 20W

A threshold comparison is meaningful only when the experimental error metric and the code's assumed error metric are actually comparable.

8.4 1 bad qubit out of 12

FABRICATION & MATERIALS · A ROOM · CALLBACK · SCIENCETANK

SCENE 40W

The twelfth qubit is found, but the weak feature beside it survives every repeat. Swapping chips would consume four of the days before Delft's slot. Lindqvist arrives with the fabrication coupon: its junction resistances spread twice as wide as usual.

GUIDE 88W

The chip has twelve qubits and eleven sit within 40 MHz of design. A warm-up and cooldown is four days out of the ten before the review. The spare comes from the same suspicious run and its coupon has not been measured. Junction resistance on the coupon spreads twice as wide as usual. The defect near 4.55 GHz has not been characterised on any chip in this laboratory. A register reported as eleven qubits is a different claim from one reported as twelve with one of them worse.

ASK 6W

Decide whether to keep this chip

BEHIND THE BUTTON 193W

Why the whole spread is graded. Funding is not a vote for one idea. A portfolio says what you think is likely, what is worth hedging against, and what is not worth pursuing at all — and the last two are where most of the information is. Backing the right proposal while quietly funding a bad one is a worse answer than it looks. That is why the small numbers count as much as the big one.

What the three numbers are for. Thirty-five is what makes a lead a lead: below it you have hedged rather than chosen. Fifteen is the most that can sit on unsupported work before it stops being a rounding error. Past that it is a second opinion nobody argued for. Twenty is the floor under a line of work you have already called strong, because funding it too thin to finish spends the money and buys nothing.

Why there is a floor on the total. Points held back are not caution; they are a decision not to decide, taken with somebody else's money and somebody else's deadline. The floor is what forces the panel to say something.

TAKES AS READ

a fabrication run produces chips with a spread of properties · energy quantisation, and why a level spacing sets everything else — taken as read

VERDICT 103W

The useful question is not whether this chip is perfect; it is whether changing chips is likely to improve the run Ridgeway is about to send to Delft. The present chip has 11 qubits close to target and a twelfth that can be driven and characterised. A swap costs four days, and the spare comes from the same suspicious fabrication run, so replacement does not remove the underlying uncertainty. The strongest plan is to keep this chip, map the nearby resonance, operate the qubit away from it if possible, and give Lindqvist the data needed to decide whether the feature tracks the process.

TAKEAWAY 19W

A device decision should compare the cost of its known imperfection with the cost and uncertainty of replacing it.

END OF THE DAY 22W

Different benchmarking procedures can produce different error numbers because they average over different operations and respond differently to preparation and measurement errors.

A number that rules something out

BEFORE THE DAY — CANVASS

Who has seen the fridge do this before

The mixing chamber has been sitting eight millikelvin warm since the last cooldown, and somebody in this building has seen that specific behaviour before. Ask along the spine until enough answers come back to say whether it is the still, a blocked line, or a thermometer that has drifted.

PLAN CARD 106W

"Tuesday, and the review wants a demonstration that the machine can produce correlations incompatible with a simple local-classical model. Anders Holm proposes a CHSH experiment: entangle two qubits, measure four combinations of settings, and compute one number. Under the Bell-test assumptions, local pre-existing outcomes obey a bound of two. Dasha Petrova insists that an on-chip experiment does not satisfy every condition of a loophole-free Bell test, and that the limitation belongs in the claim rather than in a footnote. Today you compute the quantity, identify what its value says under the stated assumptions, and separate an on-chip quantum demonstration from a loophole-free test of local realism."

BRIEFING 18W

Two qubits, four settings, and a quantity that cannot exceed two if the outcomes were agreed in advance.

WHAT YOU DECIDE 14W

Run the Bell test as a measurement, and say precisely what a violation excludes.

9.1 Four correlations, one number

ERROR & VERIFICATION · A ROOM · TALLY

SCENE 56W

Two entangled qubits, four analyser settings. Every shot returns two outcomes, and they either agree or disagree. A correlation is how often they agree minus how often they disagree, so it runs from minus one to plus one. Four of those combine into one statistic. The review deck has a blank line where that number goes.

GUIDE 77W

The trace is the statistic as the shots came in, and the question is where it converges. Early movement is the sample size, not the physics. Each row also carries its own spread, which falls as one over the square root of that row's shots, so the row with the widest spread is the one still moving the total. Twenty-four batches is the whole stop. Report the value you had when you said it had stopped moving.

ASK 24W

Acquire the four correlations yourself and combine them. How large is the statistic, and how many shots did it take before it stopped moving?

BEHIND THE BUTTON 178W

How the four go together. The statistic adds three of the correlations and subtracts one. The signs are not a convention: they are chosen so that any explanation in which each qubit carries its own predetermined answers has a hard ceiling on the total, whatever those answers are. Getting a sign wrong measures a different quantity that means nothing.

Why the number wanders at first. Each correlation is a proportion measured from a finite number of shots, and a proportion scatters around its true value by about 1 over the square root of the number of trials. Four of them combined scatter a little more. Two hundred shots gives a number that will still move; a few thousand gives one that will not.

What the comparison afterwards is for. The ceiling for any local, predetermined explanation is 2. The experiment exists to find out which side of that the measured statistic falls on, and that comparison is only worth making once the number has stopped moving — which is the decision this panel actually asks you to make.

TAKES AS READ

a correlation between two detectors can be measured as a number between minus one and one · a proportion measured from a finite number of trials scatters around its true value · superposition as an amplitude, not a mixture — taken as read

VERDICT 127W

A correlation here is not the fraction of trials that agree; it is that fraction minus the fraction that disagree, which is why it runs from -1 to $+1$ and why a pair that almost always disagrees contributes a large negative number. Every 1 of those 4 values is a proportion from a finite number of shots, so each carries its own scatter, and the scatter shrinks as the square root of the count — which is why 100 shots per

setting can put the combination either side of 600 usually does not. The bound of 2 belongs to local models satisfying the Bell assumptions; quantum mechanics allows up to $2\sqrt{2}$, about 2.83. The arithmetic is trivial. Knowing when the number has stopped moving is the measurement.

TAKEAWAY 21W

The CHSH statistic combines four measured correlations; the classical bound and quantum prediction are comparisons made after the data are combined.

9.2 What has been ruled out

NETWORKS & SECURITY · A PERSON · CHOICE

SCENE 27W

S is 2.78, comfortably above 2. Sadiq asks Holm to say in one sentence what that establishes, and then asks him to say what it does not.

GUIDE 70W

The four options differ in how much they claim the number rules out. The candidates are one class of model, every classical model, a claim about signalling, and a whole theory. Each was derived under stated assumptions, so ask what those assumptions forbid. An inequality rules out only what it was built to rule out. Claiming more is how a real result becomes the overclaim a careful reader takes apart.

ASK 14W

Assuming locality and measurement independence for the moment, what would $S = 2.78$ exclude?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a measurement can rule out a class of explanations without confirming one

VERDICT 97W

Under the CHSH assumptions, a local hidden-variable model in which outcomes are determined by pre-existing variables cannot produce $|S| > 2$. The measured value would therefore exclude that class of models. It does not imply faster-than-light signalling, and it does not prove quantum mechanics uniquely correct. On Ridgeway's chip, however, the qubits are close enough that the locality condition is not physically enforced. That means the on-chip result is a strong demonstration of nonclassical correlations and, within quantum theory, entanglement — but the

stronger loophole-free statement about local realism requires the conditions examined in the next scene.

TAKEAWAY 26W

A Bell inequality rules out a class of models only to the extent that the experiment satisfies the assumptions used to derive and test the inequality.

9.3 What a careful reader will ask

CONTROL & READOUT · A ROOM · PROTOCOL

SCENE 33W

Petrova lists what somebody sceptical will want to know: whether the settings were chosen after the state was prepared, whether the two qubits were far enough apart, and how many runs were discarded.

GUIDE 68W

The left column is four conditions a sceptic will ask about; the right is four things an apparatus physically does. Pair them by mechanism. For each condition, ask what has to be true of the equipment for the assumption to hold rather than be granted. An apparatus can enforce an assumption, or a reader can be asked to accept it. That difference separates a demonstration from a claim.

ASK 3W

State the loopholes

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

an experiment's conclusion depends on conditions its apparatus may not meet

VERDICT 91W

Ridgeway's qubits are millimetres apart and measured over microseconds, so signals travelling at or below light speed could cross between them during a trial; the locality loophole is open. Settings fixed long before the measurement also leave the measurement-independence logic weaker than a fast random choice made during each run. Outcome-dependent discarding can open a sampling loophole, and insufficient detection can require a fair-sampling assumption. An on-chip CHSH violation is still scientifically useful, but the slide should call it an on-chip

Bell/entanglement demonstration rather than a loophole-free rejection of local realism.

TAKEAWAY 19W

A loophole-free Bell claim requires the apparatus to enforce the assumptions rather than asking the reader to grant them.

9.4 How good T2 could be

CRYOGENICS & VACUUM · A ROOM · CALLBACK · BALLPARK

SCENE 35W

Okafor wants to know how much room there is before anybody spends a week chasing the noise. The relation is one over T2 equals one over twice T1 plus one over the pure dephasing time.

GUIDE 71W

Five numbers, two of which are the same relaxation time written with and without its factor of two. The relation is written in rates. A rate is the reciprocal of a time, so the factor has to sit where the algebra put it. Read the equation before reading the labels. Halving or doubling that term by mistake shifts the dephasing time enough to send somebody after noise that is not there.

ASK 8W

Estimate the pure dephasing time behind these numbers.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

two independent loss processes add as rates rather than as times · superposition as an amplitude, not a mixture — taken as read

VERDICT 69W

Under the simple exponential model, $1/T_2 = 1/(2T_1) + 1/T_\phi$. The first term is the phase loss that accompanies energy relaxation; the remainder is labelled pure dephasing. Subtracting rates gives a useful diagnostic number, not a proof of a microscopic mechanism. If the observed decay is strongly non-exponential, as can happen with slow noise, the model itself has to be revisited rather than forcing every trace into one T_ϕ .

TAKEAWAY 18W

When independent exponential decoherence processes are modelled as rates, the rates add and the corresponding times do not.

END OF THE DAY 18W

A Bell result is only as strong as the inequality, statistical test and experimental assumptions attached to it.

Reading a claim as a claim

PLAN CARD 112W

"Wednesday, and the paper everybody has been arguing about is on the table. A group has run a sampling circuit on a 67-qubit device and estimates that reproducing its output with the best classical methods they analysed would take about 9000 years. Priya Raghavan wants Ridgeway's response ready for the review. Anders Holm points out that the headline joins several different claims: what the quantum device measured, how its fidelity was inferred, and how difficult the task is for specified classical algorithms and hardware. Today you separate those claims, ask what kind of evidence supports each one, and decide what a twelve-qubit laboratory can contribute without pretending to adjudicate a sixty-seven-qubit experiment."

BRIEFING 20W

A competing group's paper says a classical computer would need 9000 years. That is a claim about a classical computer.

WHAT YOU DECIDE 12W

Work out what an advantage result asserts, and what would falsify it.

10.1 What was measured and what was estimated

ERROR & VERIFICATION · A ROOM · CASEBOOK

SCENE 28W

The paper reports a sampled distribution, a fidelity estimate, and a runtime for the best known classical algorithm on the largest machine the authors could get time on.

GUIDE 68W

Four items out of one headline, and they are not the same kind of thing. One is data the device produced. Some are estimates that lean on a model. One is conditional on what classical computers can do this year. Ask of each what would have to change for it to change. A runtime a better algorithm can overturn does not have the standing of a measured distribution.

ASK 7W

Separate the two halves of the claim

BEHIND THE BUTTON 181W

Why explanations rather than labels. Naming a finding is not accounting for it, and a plausible-sounding mechanism attached to the wrong observation is the commonest way a wrong story survives. Committing an explanation to one clue means claiming it accounts for that clue specifically and not for its neighbour, which is where the two come apart.

How to use the one-each rule. The explanations are a set to be distributed, not a list to be sampled, so every join constrains the rest. Settling the two you are confident of can decide the remaining pair by elimination. Where it does not, two explanations are still competing for one clue, and that competition is the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every explanation used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a comparison has a measured side and a modelled side

VERDICT 90W

The quantum device's sampled data are experimental observations. The fidelity estimate is partly model-dependent because at sizes beyond exact classical simulation one cannot simply calculate the full ideal distribution and compare every outcome. The quoted classical runtime is conditional on algorithms, implementation choices and available hardware; a better classical method can change that estimate without changing the quantum experiment at all. The useful reading habit is therefore to ask which part is measured, which part is inferred, and which part is a statement about the current state of competing computation.

TAKEAWAY 20W

Measured outputs, model-dependent fidelity estimates and classical-runtime estimates are different kinds of evidence even when they appear in one headline.

10.2 What 12 qubits can say about 67

CONTROL & READOUT · A PERSON · CHOICE

SCENE 22W

Ridgeway has 12 qubits. Mensah is asked what, if anything, a twelve-qubit device can contribute to an argument about a sixty-seven-qubit one.

GUIDE 69W

All four options describe something a small device could be pointed at. They differ in whether the thing being tested is the hardware, the runtime, or the method used to interpret the data. At twelve qubits, one of those has an independently computable truth to check against. Decide which, and be modest about what it settles. Validating a load-bearing method where the answer is knowable beats a scaled-up guess.

ASK 15W

Which part of the sixty-seven-qubit claim can Ridgeway test most directly with a twelve-qubit device?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a small system can be simulated exactly and a large one cannot

VERDICT 90W

"At 12 qubits, exact classical calculation of the ideal distribution is easy enough to serve as a ground truth. Ridgeway can therefore run the same *analysis method* at small size and test whether the estimator recovers the answer it is supposed to recover, including how it behaves as controlled noise is added. That does not validate the sixty-seven-qubit runtime claim by itself, nor does it prove the estimator behaves identically at all sizes. It does something more modest and useful: it tests a load-bearing method where the answer is knowable."

TAKEAWAY 17W

Small instances are valuable because they can validate an analysis against a truth that is independently computable.

10.3 The response

NETWORKS & SECURITY · A ROOM · SCIENCETANK

SCENE 29W

Raghavan wants a paragraph for the review. The options on the table run from a full rebuttal to silence, and the group has one genuinely new thing to offer.

GUIDE 85W

Ridgeway has twelve qubits against the paper's sixty-seven, and an exact classical simulation at twelve qubits takes seconds. Nobody has published a validation of this estimator against an answer that is exactly known. Two previous advantage claims were overturned by better classical algorithms rather than by experimental error. A classical runtime is therefore a statement about this year. Delft is testing a different question and its window is ten days out. The review is in three days, and silence until then is itself a position.

ASK 5W

Decide what Ridgeway says publicly

BEHIND THE BUTTON 193W

Why the whole spread is graded. Funding is not a vote for one idea. A portfolio says what you think is likely, what is worth hedging against, and what is not worth pursuing at all — and the last two are where most of the information is. Backing the right proposal while quietly funding a bad one is a worse answer than it looks. That is why the small numbers count as much as the big one.

What the three numbers are for. Thirty-five is what makes a lead a lead: below it you have hedged rather than chosen. Fifteen is the most that can sit on unsupported work before it stops being a rounding error. Past that it is a second opinion nobody argued for. Twenty is the floor under a line of work you have already called strong, because funding it too thin to finish spends the money and buys nothing.

Why there is a floor on the total. Points held back are not caution; they are a decision not to decide, taken with somebody else's money and somebody else's deadline. The floor is what forces the panel to say something.

TAKES AS READ

a public statement can be about a method rather than about a rival

VERDICT 75W

"Ridgeway cannot determine the true best classical runtime for the sixty-seven-qubit experiment with its twelve-qubit processor. What it can contribute is a validation study of the estimator: compare it with exact results at tractable sizes, inject controlled errors, and map where the estimator remains trustworthy. That evidence could support, qualify or challenge future advantage claims without requiring Ridgeway to endorse or dismiss this paper's headline. A public method is more durable than a public posture."

TAKEAWAY 22W

A constructive critique is strongest when it contributes a testable method or benchmark rather than a verdict unsupported by your own data.

10.4 What scrambles a phase without taking energy

FABRICATION & MATERIALS · A ROOM · CALLBACK · PROTOCOL

SCENE 27W

Lindqvist lists four candidates on the board and asks which of them changes a qubit's frequency slightly and slowly, rather than taking a quantum out of it.

GUIDE 69W

The four sources on the left are all real noise in this device. The four descriptions differ on one thing: whether the process takes energy out of the qubit, or only moves its frequency between repetitions. Ask that question of each source before pairing anything. A mechanism sorted into the wrong channel sends a week of shielding work at the wrong target, and the coherence number does not move.

ASK 6W

Say where slow noise comes from

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a qubit's frequency can depend on its surroundings

VERDICT 88W

Dephasing appears when the qubit accumulates a different phase from run to run. Flux noise and bias-current noise can do that by shifting the transition frequency. Residual photons in a dispersively coupled readout resonator can also shift the qubit frequency randomly and dephase it. A resonant material defect can exchange energy with the qubit and often shows up strongly in T1, although real defects can affect more than 1 coherence channel. The classification here is therefore about the dominant signature in this situation, not a universal one-to-one rule.

TAKEAWAY 23W

A process that changes the qubit frequency from one repetition to the next can wash out phase even if little energy is lost.

END OF THE DAY 16W

A computational-advantage claim combines experimental evidence with a moving statement about classical algorithms, hardware and assumptions.

The discriminator that had seen the answer

PLAN CARD 106W

Thursday, and Delft's raw shots arrived overnight. Their device has longer coherence, lower gate error and better readout separation than Ridgeway's by every independent metric they sent, yet the circuit fidelity they report is about a third lower. Anders Holm has been asking for a second device since March because this is exactly the kind of contradiction one machine cannot create for itself. Marta Castellan pulls both groups' analysis notebooks and starts comparing where calibration data end and evaluation data begin. Today you diagnose the discrepancy, trace which of the fortnight's results depend on the suspect step, and decide what has to happen before the review.

BRIEFING 18W

Delft ran the circuit. Their fidelity is a third lower than Ridgeway's, on hardware that is otherwise better.

WHAT YOU DECIDE 16W

Find out why the second device disagrees, and what that does to a fortnight of numbers.

11.1 Scoring yourself on your own training set

CONTROL & READOUT · A ROOM · HOLDOUT

SCENE 126W

When this machine reads a qubit it does not report 0 or 1. It reports a voltage. Shots that were really 0 come back low and shots that were really 1 come back high, but the two groups overlap, so no cut-off separates them perfectly. You have to choose one anyway. That cut-off is the threshold: above it you call the shot 1, below it you call it 0. A threshold's score is the share of shots it labels correctly, so 95% means 5 shots in every 100 were called wrong, and higher is better. Castellan has 800 shots whose true answers are already known, split into two trays of 400. Ridgeway picked his threshold using tray A, then reported the score tray A gave it.

GUIDE 112W

Each slider position is a different threshold, scored against tray A and plotted, so the curve is one you build by looking. Two things about those scores. Each comes from 400 shots, so it can land about a point either side of the truth, and you are hunting the best of many positions, which is a contest luck can win. And moving the line 2 mV only reclassifies the shots inside those 2 mV, so anything the clouds are really doing changes gradually. When you have chosen, freeze. Freezing commits you, and only then does tray B open — 400 shots that had no say in your choice. Report tray B's number.

ASK 14W

Choose your threshold on tray A, freeze it, and report what tray B scores.

BEHIND THE BUTTON 421W

Why there is a choice to make at all. A readout is not a clean 0 or a clean 1; it is a voltage with noise on it. Plot every shot and you get two humps, one per state, and the humps have tails that reach into each other. Put the threshold too low and true 0s get called 1. Too high and true 1s get called 0. Somewhere in between is the line that costs you least, and finding it is what this panel is for.

Why a score wobbles. Flip a fair coin 400 times and you will not get exactly 200 heads, and you will get a different count if you do it again. A percentage measured on 400 shots behaves the same way, and here that is worth about a point. So two thresholds whose scores differ by half a point have not been told apart at all.

Why the best number on the screen is not the best threshold. Sixteen positions each carry their own wobble, and you are keeping whichever came out highest. The winner is the one whose luck ran furthest in your favour, so the top score in any scan is biased upward — not because anybody cheated, but because picking the maximum is itself a choice made using the noise. Statisticians call it the winner's curse.

The discriminator is the step that applies your line and turns a voltage into a bit. Every other number in the experiment is built on top of it, so a threshold in the wrong place quietly lowers the circuit fidelity reported at the end. That is how Delft

can own the better machine and still post the worse result. Their hardware is cleaner. Ridgeway's number was flattered by the way it was measured.

Picking a rule on one set of data and scoring it on data you kept back is called a holdout. It is the standard defence against fooling yourself, and it is procedural rather than clever. You do not have to be a better statistician. You have to have sealed the second tray before you started. Note what would go wrong if you could keep adjusting after tray B opened. You would be choosing with tray B as well, and there would be nothing left to test the choice on.

The two equations attached to this card come from the course's error-correction thread, and both are spelled out below. Neither is needed here. They are on the card because the day's topic touches them.

TAKES AS READ

a rule fitted to data describes that data better than it describes new data • a labelled calibration set is a sample, and a sample has features of its own • a percentage measured on four hundred shots carries about a point of counting noise

VERDICT 161W

Two things went wrong at 14 mV and only one of them is luck. It is off the top of the curve, so it really does classify worse — about two points worse — and on top of that its tray A score was a favourable draw of 400 shots. Picking the highest of sixteen scores selects for exactly that kind of draw, which is why the top of a scan is biased upward. The middle of the curve is high on tray A for a different reason: the two voltage clouds are furthest apart there, and the next 400 shots have the same clouds. So a score measured where the threshold was chosen is not a measurement of the threshold. It is a measurement of the threshold plus the sample, biased upward every time, quietly and in the same direction. The cure is procedural rather than clever: keep shots back, freeze the rule, and report what it does on those.

TAKEAWAY 25W

A rule chosen on a sample is scored generously by that sample, so the only honest number comes from data the rule has never seen.

11.2 What the artefact touches

ERROR & VERIFICATION · A ROOM · CASEBOOK

SCENE 20W

Holm lists the numbers this fortnight produced and asks which of them passed through the discriminator and which did not.

GUIDE 68W

Four numbers, and the question is dependency rather than importance. Ask of each whether the faulty step sits upstream. Was the quantity built out of classified bits, or out of how a signal changed with delay? A fitted decay constant and a raw outcome probability inherit a readout bias very differently. Rerunning everything wastes a fortnight, and rerunning too little leaves a corrected record that is still wrong.

ASK 4W

Sort the fortnight's numbers

BEHIND THE BUTTON 181W

Why explanations rather than labels. Naming a finding is not accounting for it, and a plausible-sounding mechanism attached to the wrong observation is the commonest way a wrong story survives. Committing an explanation to one clue means claiming it accounts for that clue specifically and not for its neighbour, which is where the two come apart.

How to use the one-each rule. The explanations are a set to be distributed, not a list to be sampled, so every join constrains the rest. Settling the two you are confident of can decide the remaining pair by elimination. Where it does not, two explanations are still competing for one clue, and that competition is the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every explanation used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a fault in one step affects only what depends on that step

VERDICT 115W

"The discriminator converts analogue readout records into bits, so quantities built directly from classified outcomes — including the reported readout fidelity, circuit outcome probabilities and the Bell correlations — must be reconsidered. T1 and T2 are extracted mainly from how a signal changes with delay, so an approximately stable affine readout error often changes the amplitude or offset more than the fitted decay constant; those fits should still be checked, but they are not automatically invalid. Standard randomized benchmarking has a similar designed insensitivity to fixed state-preparation and measurement terms in its decay parameter. The rule is

dependency, not prestige: rerun or reanalyse anything for which the faulty classification can change the quantity of interest."

TAKEAWAY 23W

A pipeline fault propagates only into quantities downstream of that step, and the dependency graph matters more than how important a number feels.

11.3 Rerun, and tell whom

FABRICATION & MATERIALS · A ROOM · TRIAGE

SCENE 23W

Nothing has been published. The review is in two days. Delft has the raw shots and has not been told what Ridgeway found.

GUIDE 66W

All four actions are going to happen. The order is what is being graded, and the test is neither effort nor importance. Ask of each one which changes what somebody outside this building does in the next hour. Everything internal can be done tomorrow at the same cost. Time another group spends chasing a hardware fault that does not exist cannot be given back to them.

ASK 6W

What does the group do first?

BEHIND THE BUTTON 194W

Why first is a different question from most important. When several calls compete for the same hour, what a choice is worth is not its own importance but the difference between doing it now and doing it later. A serious problem that will be no worse in an hour costs nothing to defer. A smaller one that closes a door costs everything behind that door.

What to look for in the options. Two things separate them: which is still changing, and which is a precondition for the others. A situation that is deteriorating has a cost per hour attached to it, and a step that unblocks the rest multiplies the value of the hours after it. Everything else is a preference about where to start.

Why only one answer is marked. In the situation these options describe all of them eventually happen; what is being tested is the head of the queue, because that is where the reasoning is visible. The verdict names what each of the others was waiting on, which is worth reading even when the choice was right — the ordering behind the first place is the rest of the answer.

TAKES AS READ

a fault found before publication is a different problem from one found after

VERDICT 86W

Delft is actively analysing a disagreement whose likely cause Ridgeway has now identified. Telling them immediately lets them test the hypothesis on their raw data and stops them spending more time searching for a hardware problem that may not exist. Ridgeway should then rerun the affected metrics with a frozen or held-out evaluation set, update the review material, and audit how long the old procedure was used. The ordering matters because only the first action changes what somebody outside the building does in the next hour.

TAKEAWAY 20W

When an external collaborator is spending time on a discrepancy you already understand, disclosure becomes part of the experimental workflow.

11.4 A qubit, or the amplifier

ERROR & VERIFICATION · A PERSON · BELT

SCENE 37W

The night produced two hundred thousand shots and the fit runs at eight. Some of the structure in them is the qubit. The rest is the measurement chain, and both arrive as numbers in the same file.

GUIDE 61W

Two bins. Something the qubit did depends on what the qubit was asked to do — it moves with the drive, the delay, the frequency. Something the chain did does not: it is there when the drive is off, it tracks the room, or it sits at a fixed frequency whatever the qubit is doing. Sort on what it depends on.

ASK 11W

Send each feature to the bin that says what produced it.

BEHIND THE BUTTON 90W

Why this is the first question of every measurement. A readout chain has amplifiers, mixers, cables and a room around them, and every one of them can produce structure that looks like physics. The test is dependence: change what the qubit is doing and see whether the feature follows.

What chain artifacts look like. A feature at the same frequency whatever the qubit is tuned to. A drift that follows the room temperature. A rate that changes when somebody opens a door. Something present with the drive switched off entirely.

TAKES AS READ

a measurement passes through an instrument before it becomes a number

VERDICT 151W

Everything in the file has been through the readout chain, so the only way to tell the physics from the instrument is dependence. A real qubit feature moves when you change what the qubit is asked to do: an oscillation whose rate follows drive amplitude, a decay whose time constant is the same on repeat, a resonance that moves when the flux bias moves. Chain artifacts do not care. A spur sits at the same frequency whatever the qubit is tuned to; an amplifier's gain ripple is there with the drive switched off; a slow drift follows the room's air conditioning rather than anything on the chip. That is why the discipline is to change one thing and look, rather than to fit first and interpret afterwards — a good fit to an artifact is the most convincing wrong result a laboratory can produce, and the fit will not tell you.

TAKEAWAY 15W

A feature that does not depend on the qubit was not made by the qubit.

11.5 Copying a state is not the same as transferring one

NETWORKS & SECURITY · A ROOM · CALLBACK · CHOICE

SCENE 39W

Yusuf Sadiq points out that Delft does not need Ridgeway's physical quantum state at all; it needs the recipe for preparing its own. An arbitrary unknown state cannot be turned into a classical file that produces two perfect copies.

GUIDE 67W

All four options agree that something cannot be sent, and disagree about what. The candidates are the size of a description, the transfer itself, and what copying a recipe does. Separate the classical object here, the preparation instructions, from the physical object, the prepared state. Then ask which of the two each option is really about. Whether quantum networks are possible at all turns on that distinction.

ASK 24W

Why can Ridgeway email the preparation instructions but not email a complete classical description that lets Delft reconstruct an arbitrary unknown prepared state exactly?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

classical descriptions can be copied freely, while an arbitrary unknown quantum state cannot be perfectly cloned

VERDICT 87W

A classical preparation recipe is ordinary information and can be copied without limit. An arbitrary unknown quantum state is different: there is no universal physical operation that takes one copy and produces two perfect copies of it. That is the no-cloning theorem. It does not say quantum information can never travel; a physical quantum system can be transferred. The restriction is that the original

cannot remain behind as a second perfect copy. For this replication Delft needs the preparation instructions, not Ridgeway's particular instance of the state.

TAKEAWAY 21W

The no-cloning theorem forbids making a perfect duplicate of an arbitrary unknown quantum state; it does not forbid transferring quantum information.

END OF THE DAY 22W

A method tuned on the data it is later tested against will flatter itself, and it will do so quietly and consistently.

A day when the machine behaves

BEFORE THE DAY — EVADE

The visitor who wants the tour while the run is live

There is a funder in the building and somebody has offered them a look at the fridge. Opening the shield while a run is live loses the run, and the run is four hours in. Keep out of the way until the sequence finishes.

PLAN CARD 108W

Friday, and it is a quiet day. The fridge is at 11 millikelvin, the tune-up converged on the first attempt, every qubit is within a few megahertz of where it was yesterday, and the reruns are queued and running themselves. Kofi Mensah uses days like this to ask what the group is assuming, because it is the only time anybody has. Grace Whitfield is doing her monthly comparison against the national standard, which nobody watches and everything depends on. Today you check what a calibration actually established, read a drift that is behaving, and decide which of the group's comforts is a measurement and which is a habit.

BRIEFING 13W

Everything is calibrated, the fridge is cold, and there is nothing to fix.

WHAT YOU DECIDE 11W

Run an ordinary session, and notice what ordinary is resting on.

12.1 Converged, and on what

CONTROL & READOUT · A ROOM · CHOICE

SCENE 22W

The automatic tune-up reports success on all 12 qubits in 11 minutes. Mensah asks what, precisely, that success is a statement about.

GUIDE 68W

The four options differ in how far a converged optimiser's word is being taken. One reports on what the routine actually measured; the others extend it to drift, to next week, or to the whole device. Ask what was inside the objective function and what was never in it. The useful line is not human against automatic. It is behaviour that was tested against behaviour that was not.

ASK 16W

Which statement is justified by the tune-up's 'success' flag, and which tempting stronger claims are not?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

an automated procedure optimises the quantity it was given · resonance, detuning, and finding a qubit frequency — taken as read

VERDICT 70W

The tune-up searched a defined space of control parameters until its measured objectives — such as resonance location, pulse response and readout separation — met its convergence criteria. That is useful evidence about those objectives. It does not automatically test leakage, correlated multi-qubit errors, long-timescale drift or any other quantity outside the objective function. The important distinction is not human versus automated calibration; it is tested behaviour versus untested behaviour.

TAKEAWAY 21W

An optimizer can prove that it satisfied its objective function without proving that every relevant behaviour of the device is healthy.

12.2 The month when nothing moved

QUANTUM SENSING · A PERSON · CHOICE

SCENE 33W

Whitfield's monthly comparison agrees with the national standard to within the uncertainty, as it has for 11 months. She writes it in the log with the same care as if it had not.

GUIDE 71W

Every option is a reading of the same quiet result, and they differ in what a passing comparison is evidence for. One treats it as paperwork, one as proof, one as a statement about the reference itself. Ask what the comparison constrains, and over what interval. A month with no check is an unbounded month. The value of a quiet result is a number on how far things could have moved.

ASK 16W

What information does an in-family calibration result add to the measurements taken since the previous comparison?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a calibration can pass or fail and both are results · resonance, detuning, and finding a qubit frequency — taken as read

VERDICT 78W

If the present comparison agrees with the reference within uncertainty, then any drift of the calibrated scale over the interval is bounded at roughly that level, subject to the assumptions of the calibration schedule. That bound is what lets measurements from different months be compared on the same scale. A passing calibration does not prove perfect accuracy and does not guarantee the instrument

behaved continuously between checks; it converts an untested interval into one with a stated constraint.

TAKEAWAY 18W

A calibration that agrees with its reference is a quantitative bound on drift, not an absence of information.

12.3 Which of these is a measurement

CRYOGENICS & VACUUM · A ROOM · TRIAGE

SCENE 26W

Okafor lists four things the group is relying on today. Two are numbers somebody took this week. Two are figures nobody has revisited in a year.

GUIDE 66W

All four are things the group is relying on today, and two of them are current. Age alone does not rank them. Ask of each how many later conclusions divide by it, then weigh that reach against how long it has been since anybody looked. A stale parameter feeding one narrow interpretation is a smaller problem than a stale one every drive amplitude is computed from.

ASK 11W

Which assumption creates the largest combination of staleness and downstream leverage?

BEHIND THE BUTTON 194W

Why first is a different question from most important. When several calls compete for the same hour, what a choice is worth is not its own importance but the difference between doing it now and doing it later. A serious problem that will be no worse in an hour costs nothing to defer. A smaller one that closes a door costs everything behind that door.

What to look for in the options. Two things separate them: which is still changing, and which is a precondition for the others. A situation that is deteriorating has a cost per hour attached to it, and a step that unblocks the rest multiplies the value of the hours after it. Everything else is a preference about where to start.

Why only one answer is marked. In the situation these options describe all of them eventually happen; what is being tested is the head of the queue, because that is where the reasoning is visible. The verdict names what each of the others was waiting on, which is worth reading even when the choice was right — the ordering behind the first place is the rest of the answer.

TAKES AS READ

a figure can be current or inherited

VERDICT 85W

"The input-line attenuation is both old and load-bearing: the wiring has been rebuilt twice since it was measured, yet the value is still used to translate room-temperature power into the drive amplitude believed to reach the qubit. The fridge temperature and qubit frequencies are current. The amplifier noise temperature is less current, but it was measured on the present chain and affects a narrower part of the interpretation. Age alone is not the criterion; age multiplied by how many conclusions depend on the number is."

TAKEAWAY 18W

The riskiest inherited parameter is usually the one that is both stale and load-bearing in many later calculations.

12.4 Energy out, or phase scrambled

ERROR & VERIFICATION · A ROOM · CALLBACK · SWEEP

SCENE 62W

Two measurements run at every delay. One asks how much excitation is left in the qubit. The other asks how much phase coherence is left. Both start near one and fall as the delay grows. By convention a decay time is read where a quantity has dropped to about a third of where it started, and Holm needs the phase curve's time.

GUIDE 63W

Drag the delay. Both curves are plotted wherever you stop, and each readout gives that curve's value there. You want the delay at which the phase curve, the Ramsey contrast, has fallen to about a third of its starting value. Read it off the plot rather than working it out, then mark that delay and commit. Only the position you mark is graded.

ASK 21W

Both curves fall, and not together. At what delay has the phase coherence dropped to a third of where it started?

BEHIND THE BUTTON 209W

What the two curves are. One is energy: a qubit in the excited state eventually gives its energy away, and the excited population decays as it does. The other is phase. A qubit can keep all of its energy and still lose the phase relationship that makes it useful for computing, because the phase advances at a rate set by its frequency, and anything that shifts that frequency between repetitions scrambles it. The two are separate losses with separate cures.

Why they are read at the same place. A decay is described by the time it takes to fall to about a third of its starting value — one factor of e — and reading both curves at the same fraction is what makes the two times comparable. Read one at a third and the other at a half and the ratio between them means nothing.

Why the ratio is the point. Energy loss also destroys phase, so the phase time can never be much more than twice the energy time. If it is far shorter than that ceiling, something is scrambling phase without taking any energy — and that something is usually noise slow enough to look like a small frequency shift from one repetition to the next.

TAKES AS READ

a quantum state has both an energy and a phase, which can be lost separately · a decay time is conventionally read where a quantity has fallen to about a third of its start

VERDICT 107W

Read at the same place on both curves, the two times come out a factor of three apart, and that gap is the diagnosis. Energy is leaving slowly; something is scrambling phase three times faster without taking any energy at all, which is what slow noise shifting the qubit's frequency between repetitions does. The two have

different cures — energy loss is materials and fridge, dephasing is usually wiring — so the ratio decides who spends the week. It is also a bound worth carrying: coherence can never exceed twice the relaxation time, so this device could have reached 180 and is managing a fifth of that.

TAKEAWAY 17W

Comparing T1 with T2 tells you whether energy loss alone can explain the observed loss of coherence.

END OF THE DAY 17W

A machine that is working is a set of assumptions nobody is currently being forced to check.

Three things at once

PLAN CARD 118W

Monday, 15:20, and 3 things arrive together. A contractor with a digger has cut the fibre to the substation, which ends Yusuf Sadiq's key run mid-session. Grace Whitfield's clock comparison has come back 4×10^{-15} off, which is 10 times its usual scatter and would put every frequency in the building slightly wrong. And the review pack is due at five, with the corrected fidelity numbers still being rerun. Priya Raghavan has one group, one afternoon, and has asked for an order rather than a discussion. Today you decide what gets attention first, work out what a cut link does to a key already exchanged, and say what a drifting reference does to everything measured against it.

BRIEFING 14W

A fibre cut, a clock drifting, and a review deadline, all inside an hour.

WHAT YOU DECIDE 9W

Rank three demands on one group in one afternoon.

13.1 Which of the three cannot wait

NETWORKS & SECURITY · A ROOM · TRIAGE

SCENE 40W

Within 20 minutes, the QKD fibre goes dark, the review pack comes back with a deadline warning, and Whitfield reports a clock discrepancy. Raghavan puts all three on the board. The group can move on all three, but not equally.

GUIDE 67W

Three real problems, all urgent, and a deadline is only one kind of urgency. Ask of each whether the damage is still growing while you attend to the others, and how quickly it could be bounded. A hard deadline with a known workflow is not the same as an unbounded set of measurements quietly inheriting one error. Time to containment belongs in the ranking, not after it.

ASK 6W

What gets the group's attention first?

BEHIND THE BUTTON 194W

Why first is a different question from most important. When several calls compete for the same hour, what a choice is worth is not its own importance but the difference between doing it now and doing it later. A serious problem that will be no worse in an hour costs nothing to defer. A smaller one that closes a door costs everything behind that door.

What to look for in the options. Two things separate them: which is still changing, and which is a precondition for the others. A situation that is deteriorating has a cost per hour attached to it, and a step that unblocks the rest multiplies the value of the hours after it. Everything else is a preference about where to start.

Why only one answer is marked. In the situation these options describe all of them eventually happen; what is being tested is the head of the queue, because that is where the reasoning is visible. The verdict names what each of the others was waiting on, which is worth reading even when the choice was right — the ordering behind the first place is the rest of the answer.

TAKES AS READ

a problem still developing is different from one that has finished

VERDICT 92W

"The fibre cut has stopped the QKD link and cannot be repaired by the laboratory this afternoon; the unfinished block is not yet secret key and can be quarantined. The review pack has a hard deadline but a known workflow and reruns already in progress. The clock discrepancy is different: if the reference has drifted, every new frequency measurement made against it inherits the same scale error, so the affected set grows until the reference is checked or isolated. A repeat comparison can bound that risk quickly, which puts the clock first."

TAKEAWAY 15W

Urgency is a product of ongoing damage, downstream reach and time-to-containment, not simply deadline pressure.

13.2 What an interrupted exchange leaves behind

NETWORKS & SECURITY · A PERSON · CHOICE

SCENE 49W

The link failed while the current block was partway through sifting and before parameter estimation, error correction and privacy amplification were complete. Sadiq is asked whether the stored bits are already a key, whether they must automatically be destroyed, or whether the protocol can still process a completed subset.

GUIDE 69W

The four options differ in how much of the protocol they treat as optional. Two would use the bits in proportion to how far the exchange got, one destroys them, one waits. Ask what each later stage of the protocol is actually for, and what stays unknown if it never runs. A string of bits is not a key until something has bounded what an eavesdropper could have learned.

ASK 14W

What is the scientifically correct status of the partial data after the link fails?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

raw or sifted key material is not yet secret key; the protocol's security proof specifies how parameter estimation and privacy amplification are performed

VERDICT 104W

"The stored bits are not yet usable secret key. QKD security comes from the full authenticated protocol: parameter estimation bounds the relevant error statistics and possible information leakage, error correction reconciles the two strings, and privacy amplification shortens the result to the length justified by that bound. A fibre interruption does not create a universal law that every partial bit must be

destroyed; whether a completed subset can be processed depends on the protocol and its finite-size security rules. Ridgeway's safe action is to quarantine the raw block and use only data that can still pass the protocol's defined post-processing as a self-contained block."

TAKEAWAY 20W

Sifted or raw QKD bits become usable secret key only after the protocol has bounded leakage and completed authenticated post-processing.

13.3 Everything measured against it

QUANTUM SENSING · A ROOM · CHOICE

SCENE 31W

If the clock really has drifted, every frequency measured in the building since the last comparison is off by the same fraction — including the qubit frequencies in the review pack.

GUIDE 64W

All four options give a reason a shared drift is hard to see, and only one is about the mechanism rather than the size. Ask what the building's own cross-checks are comparing against. A common reference does not add scatter; it moves everything the same way. That is why an internal comparison can stay perfectly consistent while every number in it is displaced together.

ASK 14W

What makes a common reference drift harder to notice internally than independent instrument noise?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a measurement is a ratio against a reference · resonance, detuning, and finding a qubit frequency — taken as read

VERDICT 68W

Independent noise usually reveals itself as scatter when a measurement is repeated or compared across instruments. A shared scale error can instead push every instrument tied to the same reference in the same direction, so agreement inside the building may survive. The danger is not that the drift is necessarily larger; it is that the usual cross-checks share the same mistake. An external comparison breaks that common mode.

TAKEAWAY 19W

A shared-reference error is dangerous because it can move many instruments coherently while leaving their internal comparisons apparently consistent.

13.4 What the morning calibration is chasing

CONTROL & READOUT · A ROOM · SPOT

SCENE 42W

The morning calibration cannot do everything before the users arrive. The queue is worked to one priority, and Raghavan changes it as the day's plan changes — after the fridge cycles, after the two-qubit run is booked, after a qubit drifts out.

GUIDE 50W

Calibrate what the current priority wants and leave the rest. The priority on the rack board changes during the morning and nobody announces it. What is scored is the routines either side of a change, because they are the only ones that show whether the rack is reading the board.

ASK 13W

Calibrate to the priority on the rack board, and keep watching the board.

BEHIND THE BUTTON 78W

Why any of this drifts. Frequencies move with flux, with temperature and with two-level defects that come and go. A calibration is good for hours rather than days, which is why the morning exists at all.

Why the priority changes. On a single-qubit day it is readout, because a bad discriminator spoils everything downstream. On a two-qubit day it is the couplers. After a fridge cycle it is whatever the cycle moved, which is a different list again.

TAKES AS READ

a calibration is only good for a while

VERDICT 112W

Three priorities run across the morning and each wants a different part of the queue. Anything for readout, then anything for the two-qubit gates, then anything the fridge cycle moved. A routine can answer two at once, which is what makes the change cost real. The panel scores the window either side of each change, because most of the queue is wanted by neither priority and is correctly left for the afternoon by somebody who has read nothing. The rack's own version is the ten minutes after a two-qubit run is booked: the priority has changed to couplers, and the calibration still running is the single-qubit sweep that mattered when it started.

TAKEAWAY 15W

The cost of a withdrawn instruction is paid by whoever is still working to it.

13.5 Whose error is it

ERROR & VERIFICATION · A ROOM · CALLBACK · CASEBOOK

SCENE 37W

The first end-to-end diagnostic is in. It reports 8% wrong outcomes. Separate calibration experiments estimate about 4% readout error and less than 1% preparation error. Holm asks what Ridgeway can honestly infer about the remaining circuit contribution.

GUIDE 74W

The left column is four terms of one error budget; the right says what kind of evidence each one is. They differ in how each was got: measured on its own, seen only inside a total, or inferred once the others are modelled. Ask of each which of those three it is. A budget that quotes an inferred term as a measurement sends a pulse engineer to optimise what the readout chain is doing.

ASK 8W

Take the readout out of the qubit number

BEHIND THE BUTTON 181W

Why explanations rather than labels. Naming a finding is not accounting for it, and a plausible-sounding mechanism attached to the wrong observation is the commonest way a wrong story survives. Committing an explanation to one clue means claiming it accounts for that clue specifically and not for its neighbour, which is where the two come apart.

How to use the one-each rule. The explanations are a set to be distributed, not a list to be sampled, so every join constrains the rest. Settling the two you are confident of can decide the remaining pair by elimination. Where it does not, two explanations are still competing for one clue, and that competition is the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every explanation used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

errors from independent sources combine

VERDICT 79W

"For small, approximately independent error probabilities, adding the contributions is a useful first-order budget, but it is not an exact law. The 8% observed error therefore suggests that several per cent remain for gate and circuit effects after accounting for readout and preparation. A more careful analysis would model how the errors compose rather than simply subtracting percentages. The educational point is the separation: the number a user sees is not automatically the number a pulse engineer should optimise."

TAKEAWAY 20W

An observed error rate is only a diagnosis after you state the model used to separate preparation, gates and measurement.

END OF THE DAY 20W

Triage depends on whether a problem is still propagating, what decisions it contaminates, and how quickly it can be bounded.

The last morning it can be withdrawn

PLAN CARD 107W

Tuesday, and the abstract went out three weeks ago with the inflated fidelity in it. The conference allows corrections until five o'clock today and nothing afterwards. Priya Raghavan points out that the corrected number is still good and the talk is in four months, when the rerun will be complete anyway. Anders Holm says a number that is known to be wrong cannot be left standing on the strength of a plan to fix it later. Dasha Petrova wants to know what the correction should actually contain. Today you work out what the honest correction says, what it costs, and what the group commits to either way.

BRIEFING 14W

The inflated fidelity is in a conference abstract, and the correction window closes today.

WHAT YOU DECIDE 13W

Decide what to say about a number that is already outside the building.

14.1 What a correction is for

ERROR & VERIFICATION · A ROOM · PROTOCOL

SCENE 43W

The correction window closes today. Petrova has the draft open while the held-out rerun is still running in the next room. If Ridgeway waits for the final number, the window closes; if it files now, the correction must state exactly what is known.

GUIDE 69W

The left column is four things a correction could contain; the right says what each is worth to a reader. Pair them by asking who uses that part, and for what. A corrected number repairs one record and nothing else. The mechanism lets another group find the same fault in their own pipeline. The scope is what stops every other result in the paper falling under the same doubt.

ASK 7W

Say what the correction has to contain

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a correction is read by people deciding whether to trust the next result

VERDICT 89W

Changing only the headline number leaves other groups unable to recognise the same failure mode in their own analysis. Naming the mechanism — fitting and evaluating the discriminator on the same data — makes the correction actionable. Defining the scope matters just as much because readers need to know which Ridgeway results passed through that step and which did not. The rerun can be labelled preliminary or in progress; completeness is valuable, but waiting past a hard correction deadline would preserve a number the group already knows is biased.

TAKEAWAY 19W

A useful correction tells readers what changed, why it changed, and which conclusions the mechanism can and cannot affect.

14.2 A twelve-qubit projection from three measured qubits

CONTROL & READOUT · A PERSON · BALLPARK

SCENE 44W

"The rerun's first held-out results are in for three qubits: their single-qubit readout fidelities cluster around 0.941 rather than the 0.962 reported before. Petrova asks for a clearly labelled twelve-qubit projection under the toy assumption that every qubit has the same independent readout fidelity."

GUIDE 69W

Five numbers, and two of them are readout fidelities. One is the figure first reported and one is the corrected one. Two others belong to the gate arithmetic from an earlier day. The projection is about readout across a register, so both the quantity and its source have to match. Using the number the correction replaced makes the register look healthier and reinstates the error the rerun was for.

ASK 18W

Under that toy assumption, estimate the probability that all 12 readouts are classified correctly in the same shot.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

for this estimate only, all 12 qubits have identical independent readout fidelity of 0.941

VERDICT 82W

"If 12 readouts are independent and each is correct with probability 0.941, the probability that *all 12* are correct is $0.941^{12} \approx 0.48$. That is not a measured twelve-qubit circuit fidelity: only 3 qubits have been remeasured, the 12 qubits may have different fidelities, and readout errors can be correlated. The calculation is still useful because it shows how modest per-qubit misclassification can compound across a register — as long as the slide calls the result a projection and names the assumptions."

TAKEAWAY 16W

Exponentiating a per-qubit fidelity gives a projection only after you state the independence and identical-performance assumptions.

14.3 What each version costs

NETWORKS & SECURITY · A ROOM · CASEBOOK

SCENE 39W

Raghavan closes the conference-room door and puts two versions on the screen. One changes the headline before the correction window closes; the other leaves it standing. Before anybody signs, he wants the cost of each choice written underneath it.

GUIDE 66W

Four courses of action on the left, and on the right what each one actually costs. Pair them by consequence rather than by how the choice feels from inside the room. Silence is not the zero-cost option; it spends a window that does not reopen. Ask of each which costs are recoverable later and which are not. That asymmetry is the argument, and embarrassment is not.

ASK 6W

Decide what the group commits to

BEHIND THE BUTTON 181W

Why explanations rather than labels. Naming a finding is not accounting for it, and a plausible-sounding mechanism attached to the wrong observation is the commonest way a wrong story survives. Committing an explanation to one clue means claiming it accounts for that clue specifically and not for its neighbour, which is where the two come apart.

How to use the one-each rule. The explanations are a set to be distributed, not a list to be sampled, so every join constrains the rest. Settling the two you are confident of can decide the remaining pair by elimination. Where it does not, two explanations are still competing for one clue, and that competition is the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every explanation used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a decision not to act is also a decision with consequences

VERDICT 81W

Correcting now costs a smaller headline, an awkward explanation and the possibility that the final rerun moves the number again. Not correcting preserves a value the group already has evidence is biased and lets the correction deadline expire; Delft also has independent data that could surface the issue later. The asymmetry is not simply embarrassment versus no embarrassment. One path preserves the ability to update the scientific record while stating what is still provisional; the other deliberately gives up that option.

TAKEAWAY 21W

When both action and inaction have costs, compare their reversible and irreversible consequences rather than treating silence as the zero-cost option.

14.4 What goes out on Friday

QUANTUM SENSING · A ROOM · CALLBACK · SCIENCETANK

SCENE 44W

Elena Barros, who runs Ridgeway's sensing group and ships calibrated numbers to a hospital every month, has the least romantic view of the Delft offer: if somebody else will depend on Ridgeway's work, the method has to leave the building before the opportunity does.

GUIDE 88W

Delft's window begins in ten days and the next is a quarter away. Their device is a different design, which is the whole reason a second run counts as independent. Ridgeway's result has been measured on one device by one group. Delft has asked for sequences, not for results. Knowing the expected number before freezing an analysis changes every choice made along the way. The external review follows soon after the run. A result published first is one the second run is compared against rather than tested against.

ASK 6W

Decide what goes out on Friday

BEHIND THE BUTTON 193W

Why the whole spread is graded. Funding is not a vote for one idea. A portfolio says what you think is likely, what is worth hedging against, and what is not worth pursuing at all — and the last two are where most of the information is. Backing the right proposal while quietly funding a bad one is a worse answer than it looks. That is why the small numbers count as much as the big one.

What the three numbers are for. Thirty-five is what makes a lead a lead: below it you have hedged rather than chosen. Fifteen is the most that can sit on unsupported work before it stops being a rounding error. Past that it is a second opinion nobody argued for. Twenty is the floor under a line of work you have already called strong, because funding it too thin to finish spends the money and buys nothing.

Why there is a floor on the total. Points held back are not caution; they are a decision not to decide, taken with somebody else's money and somebody else's deadline. The floor is what forces the panel to say something.

TAKES AS READ

a result can be shared at different levels of completeness · calibration drift, and the tuning that has to happen daily — taken as read

VERDICT 81W

The Delft window is short and the point of using a different device is to create an independent test. Sending the circuit, implementation notes and stated uncertainties does not require Ridgeway to send its expected result or claim that its own number is final. Waiting until every local number is polished forfeits the independent run. Sending the expected value with the method would weaken the check, while

publishing before the cross-check turns an opportunity for pre-publication diagnosis into a post-publication dispute.

TAKEAWAY 19W

Sharing a reproducible method before the final claim is settled can create evidence without committing to the final answer.

END OF THE DAY 21W

The moment a correction stops being possible is the moment the decision needs the whole argument rather than the quickest one.

What we know, and how well

BEFORE THE DAY — TAG

Catch the fabrication lead before the etch starts

The wafer map for this batch exists on one sheet of paper, and once the etch starts nobody is opening the bay for two hours. Without the map the chips that come out are indistinguishable, and half the point of this batch is comparing two junction widths.

PLAN CARD 109W

Wednesday, and the fortnight goes in front of the review. Priya Raghavan will not put up a slide that states everything in the same typeface of certainty, because a panel that cannot tell a measurement from a projection will ask about the wrong thing and then act on the answer. Anders Holm has the reruns. Grace Whitfield wants the clock comparison in as a finding rather than a footnote. Marta Castellan wants the analysis pipeline versioned and cited. Elena Barros wants ten minutes for the thing that already works. Today you sort the fortnight's claims by the evidence under them and decide which of its habits the group keeps.

BRIEFING 16W

The review is this afternoon, and every number on the slides needs its evidence beside it.

WHAT YOU DECIDE 10W

State the fortnight's results at the confidence each has earned.

15.1 Measured, inferred, projected

ERROR & VERIFICATION · A ROOM · CASEBOOK

SCENE 23W

Holm puts four of the fortnight's statements on the board. The group believes all four. They are not held for the same reasons.

GUIDE 67W

Four statements the group believes, held for four different reasons. Sort them by what an opponent would have to do to challenge each one. The choices are repeat a measurement, run a controlled comparison, or dispute a model. Belief is not the variable here. A projection quoted with the confidence of a measurement is how a fortnight of careful work becomes a claim the group cannot defend.

ASK 6W

Sort the claims by their evidence

BEHIND THE BUTTON 181W

Why explanations rather than labels. Naming a finding is not accounting for it, and a plausible-sounding mechanism attached to the wrong observation is the commonest way a wrong story survives. Committing an explanation to one clue means claiming it accounts for that clue specifically and not for its neighbour, which is where the two come apart.

How to use the one-each rule. The explanations are a set to be distributed, not a list to be sampled, so every join constrains the rest. Settling the two you are confident of can decide the remaining pair by elimination. Where it does not, two explanations are still competing for one clue, and that competition is the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every explanation used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a claim can rest on a measurement, an inference, or a projection

VERDICT 98W

The training/evaluation flaw is directly reproducible in the pipeline and is independently exposed by the Delft comparison. The corrected 0.941 readout figure is a held-out measurement, but only on 3 qubits so far. The link between the fabrication change and the defects remains an inference from a process-window correlation until the controlled old-versus-new fabrication comparison is done. Expected behaviour under future error correction is a projection that depends on a code, a noise model and hardware performance Ridgeway has not yet reached. All four may be reasonable statements, but they ask the audience for different kinds of trust.

TAKEAWAY 18W

Evidence labels tell the audience what would have to be repeated, assumed or extrapolated to challenge each claim.

15.2 The number that goes to somebody else

QUANTUM SENSING · A PERSON · CHOICE

SCENE 31W

Barros has ten minutes and one number: a field sensitivity, going to a hospital that will act on it. She wants the uncertainty in the sentence rather than in a footnote.

GUIDE 67W

All four options are reasons an uncertainty might belong in the sentence, and they differ in who it is for. One is about tone, one about the number looking worse, one about completeness. Ask instead what the hospital has to decide tomorrow with this figure in front of it. Without a noise floor they cannot tell a change in a patient from a change in the instrument.

ASK 20W

Why is a sensitivity or field result incomplete for the hospital if the associated uncertainty or noise floor is omitted?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a delivered number is used by somebody who cannot check it

VERDICT 68W

The hospital needs to decide whether a change in its observed signal is larger than what Ridgeway's instrument could plausibly produce through measurement noise and calibration uncertainty. A bare central value or sensitivity number does not supply that comparison. The associated uncertainty, noise floor and calibration status are therefore operational information, not ceremonial caution. They tell the user how large a difference must be before it deserves interpretation.

TAKEAWAY 17W

A figure handed to somebody else has to carry enough with it for them to judge it.

15.3 What outlives the fortnight

CRYOGENICS & VACUUM · A ROOM · TRIAGE

SCENE 25W

Four things the group started doing under pressure. Okafor asks which of them are worth their cost on an ordinary Tuesday, when nothing is wrong.

GUIDE 69W

All four practices are good ones and all would survive an ordinary Tuesday. They differ in whether they block the failure that actually happened, or merely help reconstruct it afterwards. Trace each back to the mechanism: a classifier judged on the data it was tuned on. A practice that makes a failure legible in hindsight is worth keeping, and it is not the same as one that prevents it.

ASK 14W

Which permanent practice most directly prevents the discriminator mistake from recurring at low cost?

BEHIND THE BUTTON 194W

Why first is a different question from most important. When several calls compete for the same hour, what a choice is worth is not its own importance but the difference between doing it now and doing it later. A serious problem that will be no worse in an hour costs nothing to defer. A smaller one that closes a door costs everything behind that door.

What to look for in the options. Two things separate them: which is still changing, and which is a precondition for the others. A situation that is deteriorating has a cost per hour attached to it, and a step that unblocks the rest multiplies the value of the hours after it. Everything else is a preference about where to start.

Why only one answer is marked. In the situation these options describe all of them eventually happen; what is being tested is the head of the queue, because that is where the reasoning is visible. The verdict names what each of the others was waiting on, which is worth reading even when the choice was right — the ordering behind the first place is the rest of the answer.

TAKES AS READ

a practice adopted in a crisis has a cost that only shows later

VERDICT 83W

The failure survived because a fitted classifier was judged on the same data used to tune it. Requiring a genuinely held-out evaluation set for fitted predictive or classification steps directly blocks that failure mode and costs only extra shots and a few minutes. Versioning the pipeline remains valuable because it makes later reconstruction possible, but it does not by itself prevent optimistic evaluation. Repeating every experiment doubles cost without targeting this mechanism, and a daily crisis meeting is unlikely to survive ordinary operations.

TAKEAWAY 22W

Keep inexpensive practices that directly block a demonstrated failure mode; distinguish them from good documentation that only helps reconstruct the failure afterwards.

15.4 How long a flip takes

CONTROL & READOUT · A ROOM · CALLBACK · BALLPARK

SCENE 23W

With the drive at 4.61 GHz, the qubit's population oscillates as the pulse gets longer. Mensah measures a full cycle every 84 nanoseconds.

GUIDE 77W

Five numbers and only two of them describe the oscillation. The rest are the carrier frequency, the temperature and the size of the chip. All are true of the device; none is about how long a rotation takes. Ask of each whether it describes the population moving or the machine it moves in. Confusing the drive frequency with the rate the population turns at is the standard error here, and the two differ by orders of magnitude.

ASK 10W

Estimate the length of a pulse that flips this qubit.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a driven two-level system cycles between its states while the drive is on ·
superposition as an amplitude, not a mixture — taken as read · resonance, detuning,
and finding a qubit frequency — taken as read

VERDICT 72W

"Driving on resonance does not simply leave the qubit excited. Under a coherent drive the population oscillates between the two basis states at the Rabi frequency. A π pulse corresponds to half of 1 population cycle, so the 84 ns period gives a first calibration near 42 ns. That number is only a starting point: changing drive amplitude changes the Rabi frequency, and strong drives can introduce leakage or other off-resonant effects."

TAKEAWAY 17W

A resonant drive produces an oscillation; the pulse area, not simply the carrier frequency, determines the rotation.

END OF THE DAY 19W

A strong review separates direct measurements, model-dependent inferences and forward projections instead of giving them the same visual certainty.

THE CLOSING CARD

The correction went out at twenty past four, naming the mechanism as well as the number, and two other groups wrote within the month to say they had found the same thing in their own tune-up. Delft ran the circuit again on a held-out analysis and the two devices agree now, at a figure lower than the one Ridgeway first reported and higher than the one that frightened everybody in the middle of the fortnight.

What it cost: a headline number reduced by a third, a review spent explaining a mistake, and four days of fridge time nobody got back. What is unfinished: the oxidation change is still uncontrolled, the defect map covers three chips out of fourteen, the gate error is above the threshold where error correction starts to help, and the habit that would have caught all of it — evaluate every fitted analysis on data it has never seen — is a line in a report until the next person makes it a rule.

Card text only — no options, boards or answers. 14,542 words of card prose across 61 stops. Generated from themes/quantum by tools/make-cardbook.mjs.